Riemannian Geometry

Instructor: John Lott
Scribe: cc

Contents

1 Lecture 1 is omitted. **4**

2 Lecture 2 **4**
2.1 Flow of a Vector Field . . . 4
2.2 The Lie Derivative . . . 4
2.3 Cartan's Magic Formula . . . 5
2.4 Lie Derivative of Vector Fields . . . 5

3 Lecture 3 **7**
3.1 Riemannian Metric . . . 7
3.2 Isometries . . . 7
3.3 Examples of Constructions . . . 7
3.4 Vector Bundles and Distributions . . . 8
3.5 Riemannian Submersion . . . 8
3.6 Riemann Covering Spaces . . . 9

4 Lecture 4 **10**
4.1 Raising and Lowering Indices (Musical Isomorphisms) . . . 10
4.2 Volume Form and Density . . . 10
4.3 Length and Distance . . . 11
4.4 Maximally Symmetric Spaces . . . 11
4.5 Examples of Riemannian Manifolds with Maximal Symmetry . . . 12

5 Lecture 5 **13**
5.1 Three Models of Hyperbolic Space $\mathbb{H}^n$ . . . 13
5.2 Isometry Groups . . . 13
5.3 Geometric Schwarz Lemma . . . 14
5.4 Connections on Vector Bundles . . . 14

6 Lecture 6 **15**
6.1 The Curvature Tensor . . . 15
6.2 Linear Connections in Local Coordinates . . . 15
6.3 Induced Connections . . . 15

7 Lecture 7 **17**
7.1 Geodesics . . . 17
7.2 Parallel Transport . . . 17
7.3 Connections on Submanifolds of $\mathbb{R}^n$ . . . 17

8 Lecture 8 **19**
8.1 Holonomy Group . . . 19
8.2 Metric Compatibility . . . 19
8.3 Torsion and Levi-Civita Connection . . . 20
8.4 Invariant Metrics and Compact Groups . . . 20

9 Lecture 9 **21**
9.1 Holonomy Groups . . . 21
9.2 Levi-Civita Connection and Christoffel Symbols . . . 21
9.3 Properties of Geodesics . . . 21
9.4 Isometries and Geodesics . . . 21
9.5 The Exponential Map . . . 22

10 Lecture 10 **24**
10.1 Exponential Map and Normal Coordinates . . . 24
10.2 Uniformly Normal Neighborhoods . . . 24
10.3 Geodesic as length-minimizing curves . . . 25

11 Lecture 11 **27**
11.1 Isometries and Geodesics . . . 27
11.2 The First Variation Formula . . . 27
11.3 Geodesic Polar Coordinates and Gauss Lemma . . . 29
11.4 Geodesics are Locally Length-Minimizing . . . 29

12 Lecture 12 **31**
12.1 Geodesics as Locally Minimizing Curves . . . 31
12.2 Geodesic Completeness and the Hopf-Rinow Theorem . . . 31
12.3 Covering Maps and Local Isometries . . . 32

13 Lecture 13 **34**
13.1 Distance function . . . 34
13.2 Riemann Covering . . . 35

14 Lecture 14 **36**
14.1 Riemann Curvature Tensor . . . 36
14.2 Sectional Curvature & Schur's Theorem . . . 37
14.3 Isometric Immersions . . . 38
14.4 Induced Connection on Submanifolds . . . 38

15 Lecture 15 **40**
15.1 Proof of Symmetries of Curvature Tensors . . . 40
15.2 Cartan's Moving Frame . . . 41
15.3 Submanifolds and the Second Fundamental Form . . . 41
15.4 Hypersurfaces in Euclidean Space . . . 43

16 Lecture 16 **44**
16.1 Shape Operator and Curvature of Hypersurfaces . . . 44
16.2 Theorema Egregium and Sectional Curvature . . . 45
16.3 Relation Between Curvature and Geodesics via Jacobi Fields . . . 47

17 Lecture 17 **49**
17.1 Jacobi Fields 49
17.2 Constant Sectional Curvature and Polar Coordinates 50
17.3 Conjugate Points 50

18 Lecture 18 **52**
18.1 The Cartan-Hadamard Theorem 52
18.2 2nd Variation of Length 52

19 Lecture 19 **55**
19.1 Classical Theorems 55
19.2 The Cut Locus 57

20 Lecture 20 **58**
20.1 Injectivity Domain and Injectivity Radius 58
20.2 Comparison Geometry 58
20.3 Matrix Riccati Equation Comparison 60

1 Lecture 1 is omitted.

2 Lecture 2

Tensor Fields and $C^\infty(M)$-Multilinear Maps

Setup: Consider a tensor field $F \in \mathcal{T}_l^k M$. Let $\omega^1, \ldots, \omega^k \in \Omega^1(M)$ be 1-forms and $X_1, \ldots, X_l \in \mathfrak{X}(M)$ be vector fields.

We can define a map:

$$\mathcal{F} : \underbrace{\Omega^1(M) \times \cdots \times \Omega^1(M)}_{k} \times \underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{l} \to C^\infty(M)$$

by the pointwise evaluation:

$$\mathcal{F}(\omega^1, \ldots, \omega^k, X_1, \ldots, X_l)(p) = F(\omega^1(p), \ldots, \omega^k(p), X_1(p), \ldots, X_l(p))$$

Clearly, $\mathcal{F}$ is $C^\infty(M)$-multilinear.

Lemma 2.1. *Any $C^\infty(M)$-multilinear map $\mathcal{F}$ comes from a unique tensor field $F \in \mathcal{T}_l^k M$.*

- **Algebraic Result:** For finite-dimensional vector spaces, we have the isomorphism $(V \otimes \cdots \otimes V)^* \cong V^* \otimes \cdots \otimes V^*$. In the context of manifolds, this extends to the module level over $C^\infty(M)$.

2.1 Flow of a Vector Field

Definition 2.2. Let $X \in \mathfrak{X}(M)$ be a vector field. The *flow* of X is a collection of (local) diffeomorphisms $\{\Phi_t\}$ such that if $\gamma(t) = \Phi_t(p)$, then:

$$\frac{d\gamma}{dt} = X(\gamma(t)), \quad \gamma(0) = p$$

Definition 2.3. X is called *complete* if the flow $\Phi_t(p)$ exists for all $p \in M$ and all $t \in \mathbb{R}$. In this case, $\{\Phi_t\}$ forms a 1-parameter group of diffeomorphisms: $\Phi_{s+t} = \Phi_s \circ \Phi_t$.

Proposition 2.4. *If M is a compact manifold, then every vector field X on M is complete.*

(Note: From now on, we assume vector fields are complete for simplicity.)

2.2 The Lie Derivative

Lie Derivative of Functions

Definition 2.5. For $f \in C^\infty(M)$, the Lie derivative of f along X is:

$$\mathcal{L}_X f = \left.\frac{d}{dt}\right|_{t=0} (\Phi_t^* f) = \left.\frac{d}{dt}\right|_{t=0} (f \circ \Phi_t) = Xf$$

Lie Derivative of Differential Forms

Definition 2.6. For $\omega \in \Omega^k(M)$, the Lie derivative is:

$$\mathcal{L}_X \omega = \left.\frac{d}{dt}\right|_{t=0} \Phi_t^* \omega$$

Lemma 2.7. *The Lie derivative commutes with the exterior derivative:* $\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega)$.

Proof. Since the pull-back Φ_t^* commutes with d:

$$\mathcal{L}_X(d\omega) = \frac{d}{dt}\bigg|_{t=0} \Phi_t^*(d\omega) = \frac{d}{dt}\bigg|_{t=0} d(\Phi_t^*\omega) = d\left(\frac{d}{dt}\bigg|_{t=0} \Phi_t^*\omega\right) = d(\mathcal{L}_X\omega)$$

□

Interior Multiplication

Define the operator $i_X : \Omega^k(M) \to \Omega^{k-1}(M)$ by:

$$(i_X\omega)(X_1, \ldots, X_{k-1}) = \omega(X, X_1, \ldots, X_{k-1})$$

i_X is a graded derivation of degree -1:

$$i_X(\omega \wedge \eta) = (i_X\omega) \wedge \eta + (-1)^{\deg \omega}\omega \wedge (i_X\eta)$$

2.3 Cartan's Magic Formula

Lemma 2.8. *The algebra* $\Omega^*(M)$ *is generated by* $C^\infty(M)$ *and* $\{df \mid f \in C^\infty(M)\}$.

Proof. Choose a coordinate neighborhood (U, x^i). Any $\omega \in \Omega^k(U)$ can be written as $\omega = \sum \omega_I dx^I$. Using a partition of unity $\sum \phi_i = 1$ subordinate to an atlas $\{U_i\}$, we can write any global form as a sum of such local terms. □

Proposition 2.9 (Cartan's Magic Formula)**.** $\mathcal{L}_X = di_X + i_X d$

Proof. Both $\mathcal{L}_X$ and $di_X + i_X d$ are graded derivations of degree 0 on $\Omega^*(M)$. Thus, it suffices to check the identity on generators:

1. **On functions** $f \in C^\infty(M)$**:**

$$(di_X + i_X d)f = d(i_X f) + i_X(df) = 0 + df(X) = Xf = \mathcal{L}_X f$$

2. **On exact 1-forms** df**:**

$$(di_X + i_X d)df = d(i_X df) + i_X(d^2 f) = d(Xf) + 0 = d(\mathcal{L}_X f) = \mathcal{L}_X(df)$$

Since the operators agree on generators and satisfy the same Leibniz rule, they are equal. □

2.4 Lie Derivative of Vector Fields

Definition 2.10. Let $Y \in \mathfrak{X}(M)$. Its Lie derivative along X is defined as:

$$\mathcal{L}_X Y = \frac{d}{dt}\bigg|_{t=0} (\Phi_{-t})_* Y = \frac{d}{dt}\bigg|_{t=0} (\Phi_t^* Y)$$

Proposition 2.11. *Let* ω *be a 1-form and* X, Y *be vector fields. Then:*

$$X(\omega(Y)) = (\mathcal{L}_X\omega)(Y) + \omega(\mathcal{L}_X Y)$$

Proof. Evaluating at $p \in M$:

$$\begin{aligned}
X(\omega(Y))_p &= \frac{d}{dt}\bigg|_{t=0} (\omega(Y))(\Phi_t(p)) \\
&= \frac{d}{dt}\bigg|_{t=0} \langle \Phi_t^* \omega, \Phi_t^* Y \rangle_p \\
&= \langle \frac{d}{dt}\bigg|_{t=0} \Phi_t^* \omega, Y \rangle_p + \langle \omega, \frac{d}{dt}\bigg|_{t=0} \Phi_t^* Y \rangle_p \\
&= (\mathcal{L}_X \omega)(Y) + \omega(\mathcal{L}_X Y)
\end{aligned}$$

□

Proposition 2.12. $\mathcal{L}_X Y = [X, Y]$, *where* $[X, Y]$ *is the Lie bracket.*

Proof. Let $f \in C^\infty(M)$ be a test function. Consider the 1-form df:

$$\begin{aligned}
\langle df, \mathcal{L}_X Y \rangle &= X(df(Y)) - (\mathcal{L}_X df)(Y) \quad \text{(using the previous Prop)} \\
&= X(Yf) - d(Xf)(Y) \\
&= X(Yf) - Y(Xf) = [X, Y]f
\end{aligned}$$

Since this holds for all f, $\mathcal{L}_X Y = [X, Y]$. □

3 Lecture 3

3.1 Riemannian Metric

Definition 3.1. A **Riemannian metric** on a smooth manifold M is a covariant tensor field g of type $(0,2)$ which is symmetric and positive definite. Specifically, for each $p \in M$, $g_p : T_pM \times T_pM \to \mathbb{R}$ satisfies:

1. **Symmetry:** $g(X,Y) = g(Y,X)$ for all $X, Y \in \mathfrak{X}(M)$.
2. **Positive Definiteness:** $g(X,X) \geq 0$, and $g(X,X) = 0 \iff X = 0$.

Notation: We often write $\langle X, Y \rangle = g(X,Y)$.

Proposition 3.2. *Every smooth manifold M admits a Riemannian metric.*

Sketch of Proof. Take an atlas $\{U_i\}$ of M. On each U_i, define a local metric $g_i = \sum (dx^k)^2$. Let $\{\phi_i\}$ be a partition of unity subordinate to $\{U_i\}$. Then $g = \sum \phi_i g_i$ is a well-defined global Riemannian metric because a convex combination of positive definite symmetric forms remains positive definite and symmetric. □

Suppose we are given a local coordinate $(U, \{x^1, ..., x^n\})$, then we can expressed the metric locally as $g = g_{ij} dx^i dx^j$ since g is symmetric. Further, if M has a local frame $\{E_1, ..., E_n\}$ with coframe $\{\varepsilon^1, ..., \varepsilon^n\}$, then $g = \sum g(E_i, E_j)\varepsilon^i \varepsilon^j$. We can consider the case where $\{E_i\}$ is orthonormal, in this case, $g = \sum (\varepsilon^i)^2$.

Proposition 3.3 (Existence of Orthonormal Frame)**.** *If $\{X_j\}$ is a local frame on (M,g), then there exists an orthonormal frame $\{E_i\}$ obtained from $\{X_j\}$ via the Gram-Schmidt process.*

3.2 Isometries

Definition 3.4. Let (M,g) and $(\tilde{M}, \tilde{g})$ be Riemannian manifolds.

1. A diffeomorphism $\Phi : M \to \tilde{M}$ is an **isometry** if $g = \Phi^* \tilde{g}$, i.e.,
$$g_p(X,Y) = \tilde{g}_{\Phi(p)}(d\Phi_p(X), d\Phi_p(Y)) \quad \forall X, Y \in T_pM$$
2. The set of all isometries from (M,g) to itself forms the **Isometry Group**, denoted $\mathrm{Isom}(M,g)$.

3.3 Examples of Constructions

1. Induced Metric: If $i : M \to \tilde{M}$ is an immersion and $\tilde{g}$ is a metric on $\tilde{M}$, then $g = i^* \tilde{g}$ is the induced Riemannian metric on M.
2. Warped Product: Given $(M_1, g_1), (M_2, g_2)$ and $f \in C^\infty(M_1, \mathbb{R}^+)$, the warped product $M_1 \times_f M_2$ has the metric:
$$g = g_1 + f^2 g_2$$
Formally: $g((X_1, Y_1), (X_2, Y_2)) = g_1(X_1, X_2) + f^2(p_1) g_2(Y_1, Y_2)$.

Example 3.5 (Warped Product)**.** Here are some examples of warped products:

- The polar coordinates on $\mathbb{R}^2 \setminus \{0\}$ can be viewed as a warped product $(0, \infty) \times_r S^1$ with metric $dr^2 + r^2 d\theta^2$.
- The sphere $S^n \setminus \{\text{north, south poles}\}$ can be represented as $(0, \pi) \times_{\sin r} S^{n-1}$ with metric $dr^2 + \sin^2(r) g_{S^{n-1}}$.

Riemann Submanifolds

Lemma 3.6. *Let $i : M \to \tilde{M}$ be an embedding and $\tilde{g}$ a Riemannian metric on $\tilde{M}$. Then the induced metric $g = i^*\tilde{g}$ makes (M, g) a Riemannian submanifold of $(\tilde{M}, \tilde{g})$.*

Riemann Products

Given two Riemannian manifolds (M_1, g_1) and (M_2, g_2), the product manifold $M_1 \times M_2$ has a natural Riemannian metric defined by:

$$g((X_1, Y_1), (X_2, Y_2)) = g_1(X_1, X_2) + g_2(Y_1, Y_2)$$

3.4 Vector Bundles and Distributions

Let $E \to M$ be a vector bundle. An **inner product on** E is a smooth choice of inner product $\langle \cdot, \cdot \rangle$ on each fiber E_m. For any smooth sections $s_1, s_2 \in \Gamma(E)$, the function $m \mapsto \langle s_1(m), s_2(m) \rangle$ is in $C^\infty(M)$.

Vertical and Horizontal Bundles: Let $\pi : \tilde{M} \to M$ be a submersion.

- **Vertical Bundle:** $T^V\tilde{M} = \ker(d\pi)$.
- **Splitting:** A choice of **Horizontal Distribution** $T^H\tilde{M}$ such that $T\tilde{M} = T^V\tilde{M} \oplus T^H\tilde{M}$.
- **Riemannian Choice:** If $\tilde{M}$ has metric $\tilde{g}$, we naturally define $T^H\tilde{M} = (T^V\tilde{M})^\perp$.

Proposition 3.7. *Let $\pi : \tilde{M} \to M$ be smooth submersion of Riemannian manifolds. Then*

- *Every smooth vector field W on $\tilde{M}$ can be uniquely decomposed as $W = W^V + W^H$ with $W^V \in T^V\tilde{M}$ and $W^H \in T^H\tilde{M}$.*
- *For any smooth vector field X on M, there exists a unique horizontal lift $\tilde{X}$ on $\tilde{M}$ such that $d\pi(\tilde{X}) = X$.*

3.5 Riemannian Submersion

Definition 3.8. A submersion $\pi : (\tilde{M}, \tilde{g}) \to (M, g)$ is a **Riemannian submersion** if for all $\tilde{p} \in \tilde{M}$, the map $d\pi_{\tilde{p}}$ restricted to the horizontal space is a linear isometry:

$$d\pi_{\tilde{p}} : (T^H_{\tilde{p}}\tilde{M}, \tilde{g}) \xrightarrow{\cong} (T_{\pi(\tilde{p})}M, g)$$

Recall that we say a group acts **freely** on X if $gX = X$ implies $g = e$. We say the action is **proper** if the map $G \times \tilde{M} \to \tilde{M} \times \tilde{M}$, $(g, p) \mapsto (gp, p)$ is a proper map (i.e. the preimage of any compact set is compact).

Proposition 3.9. *Let G be a Lie group acting freely and properly on $(\tilde{M}, \tilde{g})$ by isometries. Then the quotient $M = \tilde{M}/G$ is a smooth manifold and admits a unique Riemannian metric g such that the projection $\pi : \tilde{M} \to M$ is a Riemannian submersion.*

Proof. First, what is $\tilde{M}/G$? This is the set of orbits and the topology is the quotient topology. We define the Riemann metric on M as follows: For $p \in M$, choose representative $\tilde{p} \in \tilde{M}$ with $\pi(\tilde{p}) = p$. For vectors u, $v \in T_pM$, define

$$g(u, v) = \tilde{g}(\tilde{u}, \tilde{v}) \quad \tilde{u},\ \tilde{v} \in T^H_{\tilde{p}}\tilde{M} \text{ so that } d\pi_{\tilde{p}}(\tilde{u}) = u \text{ and } d\pi_{\tilde{p}}(\tilde{v}) = v.$$

We have to check this is well-defined. If we choose another representative $\tilde{p}' = g\tilde{p}$ for some $g \in G$, then since G acts by isometries, we have

$$\tilde{g}((dg)_{\tilde{p}}(\tilde{u}), (dg)_{\tilde{p}}(\tilde{v})) = \tilde{g}(\tilde{u}, \tilde{v}).$$

Also, since $\pi \circ g = \pi$, we have $(dg)_{\tilde{p}}$ maps horizontal spaces to horizontal spaces because it is an isometry. Thus the definition of g is independent of the choice of representative.

Finally, smoothness of g follows from the smoothness of $\tilde{g}$ and the local triviality of the bundle structure induced by the free and proper action. $\square$

3.6 Riemann Covering Spaces

Definition 3.10. Let $\pi : \tilde{M} \to M$ be a smooth map. We say π is a **covering map** if for every $m \in M$, there exists an open neighborhood $U \subset M$ of m such that:

$$\pi^{-1}(U) = \coprod_i V_i$$

where each $V_i \subset \tilde{M}$ is an open set, and the restriction $\pi|_{V_i} : V_i \to U$ is a **diffeomorphism** for every i.

Metrics on Covering Spaces

If g is a Riemannian metric on the base manifold M, we can naturally obtain a Riemannian metric $\tilde{g}$ on the covering space $\tilde{M}$ via the pullback map:

$$\tilde{g} = \pi^* g$$

Since π is a local diffeomorphism, $\tilde{g}$ is a well-defined Riemannian metric on $\tilde{M}$, making π a local isometry.

*Normal Coverings and Descent of Metrics

Now consider the inverse problem: given a Riemannian metric $\tilde{g}$ on $\tilde{M}$, when does it descend to a metric g on M such that $\pi^* g = \tilde{g}$?

Definition 3.11. $\pi : \tilde{M} \to M$ is a **normal (or Galois) covering** if the fundamental group $\pi_1(\tilde{M})$ is a normal subgroup of $\pi_1(M)$. In this case, the group of Deck transformations is:

$$G = \mathrm{Deck}(\tilde{M}/M) \cong \pi_1(M)/\pi_1(\tilde{M})$$

Proposition 3.12. *Suppose $\pi : \tilde{M} \to M$ is a normal covering and $\tilde{g}$ is a Riemannian metric on $\tilde{M}$. There exists a unique Riemannian metric g on M such that $\pi^* g = \tilde{g}$ if and only if the group G acts on $(\tilde{M}, \tilde{g})$ by **isometries**.*

4 Lecture 4

4.1 Raising and Lowering Indices (Musical Isomorphisms)

The Riemannian metric g provides a natural isomorphism between the tangent bundle TM and the cotangent bundle T^*M.

Lowering Indices (The Flat Operator $\flat$)

Let $X \in T_pM$. We define $X^\flat \in T_p^*M$ by:

$$X^\flat(Y) = \langle X, Y \rangle_g = g(X, Y)$$

In local coordinates, if $X = X^j \partial_j$, then:

$$X^\flat(\partial_i) = g(\partial_j, \partial_i) X^j = g_{ij} X^j$$

Thus, the component expression is:

$$X^\flat = (g_{ij} X^j) dx^i = X_i dx^i$$

Raising Indices (The Sharp Operator $\sharp$)

Let $\omega \in T_p^*M$. We define $\omega^\sharp \in T_pM$ by the condition:

$$\langle \omega^\sharp, X \rangle_g = \omega(X) \quad \forall X \in T_pM$$

In local coordinates, if $\omega = \omega_i dx^i$, then:

$$\omega^\sharp = (g^{ij} \omega_i) \partial_j = \omega^j \partial_j$$

where (g^{ij}) is the inverse matrix of (g_{ij}), i.e., $g^{ij} g_{jk} = \delta^i_k$.

Examples

1. **Gradient:** For $f \in C^\infty(M)$, the differential $df = \partial_i f dx^i$ is a 1-form. Its "sharp" version is the gradient vector field:
$$\text{grad} f = (df)^\sharp = g^{ij} \partial_i f \partial_j$$
2. **Inner Product of 1-forms:** Let $\omega, \eta \in \Omega^1(M)$. Their inner product is defined via the isomorphism:
$$\langle \omega, \eta \rangle_g = \langle \omega^\sharp, \eta^\sharp \rangle_g = g^{ij} \omega_i \eta_j$$

4.2 Volume Form and Density

Let (M, g) be an oriented Riemannian manifold of dimension n.

Definition 4.1. The **Riemannian volume form** dV is the unique n-form such that for any oriented orthonormal basis $\{E_1, \ldots, E_n\}$ of T_pM, we have:

$$dV(E_1, \ldots, E_n) = 1$$

Proposition 4.2. *In local coordinates $(x^1, \ldots, x^n)$, the volume form is expressed as:*

$$dV = \sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$$

*The corresponding measure for integration is often denoted as the **Riemannian density**:*

$$d\mu = \sqrt{\det g}\, d^n x$$

4.3 Length and Distance

Let $\gamma : [a, b] \to M$ be a smooth curve.

- **Length:** The length of γ is defined as:

$$L(\gamma) = \int_a^b |\dot{\gamma}(t)|_g \, dt = \int_a^b \sqrt{g(\dot{\gamma}(t), \dot{\gamma}(t))} \, dt$$

- **Invariance:** Length is invariant under reparametrization.
- **Regularity:** γ is *regular* if $\dot{\gamma}(t) \neq 0$ for all t (i.e., it is an immersion). γ is *admissible* if it is continuous and piecewise regular.
- **Arc-length Parameter:** $s(t) = \int_a^t |\dot{\gamma}(u)| du$. Under this reparametrization, $|\frac{d\gamma}{ds}| = 1$ (unit speed).

Proposition 4.3. *If $(M, g) \to (\tilde{M}, \tilde{g})$ is an isometry, then the induced map preserves lengths. i.e. $L(\gamma) = L(\Phi \circ \gamma)$.*

Distance Definition: For a connected Riemannian manifold (M, g), the distance $d(p, q)$ between $p, q \in M$ is defined as:

$$d(p, q) = \inf\{L(\gamma) \mid \gamma : [a, b] \to M \text{ is admissible}, \gamma(a) = p, \gamma(b) = q\}$$

This definition exists since the manifold is connected and the path is compact.

Theorem 4.4. *(M, d) is a metric space, and the metric topology coincides with the manifold topology.*

The key step is to show that in local coordinates, the Riemannian metric is equivalent to the Euclidean metric.

Lemma 4.5. *Let (M, g) be Riemannian manifold and $(U, \{x^1, ..., x^n\})$ be a local coordinate system. Set $\bar{g}$ be the Euclidean metric on U defined by $\bar{g} = \sum (dx^i)^2$. Then for any compact set $K \subset U$, and $x \in K$, $v \in T_x M$, we have*

$$|v|_{\bar{g}} \approx_{C_k} |v|_g.$$

Proof. Consider the continuous function $F : TU \to \mathbb{R}$ by $F(x, v)|v|_g$. Since K is compact, F attains a minimum and maximum on the set $\{(x, v) \in TU | x \in K, |v|_{\bar{g}} = 1\}$. □

4.4 Maximally Symmetric Spaces

Definitions

- **Homogeneous:** The isometry group $\mathrm{Isom}(M, g)$ acts transitively on M.
- **Isotropic:** The isometry group $\mathrm{Isom}(M, g)$ acts transitively on the unit tangent bundle $T^1 M$. Recall the action is given by $d\Phi_p(v)$ for $\Phi \in \mathrm{Isom}(M, g)$, $p \in M$, and $v \in T_p^1 M$.
- **Frame Isotropic:** $\mathrm{Isom}(M, g)$ acts transitively on the bundle of orthonormal frames $O(M)$. This implies M has constant sectional curvature.

4.5 Examples of Riemannian Manifolds with Maximal Symmetry

1. **Euclidean Space $\mathbb{R}^n$:** Standard metric $g = \sum (dx^i)^2$. $\mathrm{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$. $\dim = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.

2. **Sphere S^n:** Metric induced from $\mathbb{R}^{n+1}$. $\mathrm{Isom}(S^n) = O(n+1)$. $\dim = \frac{n(n+1)}{2}$.

3. **Hyperbolic Space $\mathbb{H}^n$:** Constant curvature $K = -1$.

   - **Hyperboloid Model:** In $\mathbb{R}^{n,1}$, surface $\tau^2 - \sum (\xi^i)^2 = 1, \tau > 0$.
   - **Poincaré Ball Model:** $B^n = \{x \in \mathbb{R}^n \mid |x| < 1\}$, $g = \frac{4\sum (dx^i)^2}{(1-|x|^2)^2}$.
   - **Upper Half-Plane Model:** $U^n = \{(x_1, \ldots, y) \in \mathbb{R}^n \mid y > 0\}$, $g = \frac{\sum (dx^i)^2 + dy^2}{y^2}$.

   $\mathrm{Isom}(\mathbb{H}^n) \cong O(n,1)$. $\dim = \frac{n(n+1)}{2}$.

5 Lecture 5

5.1 Three Models of Hyperbolic Space $\mathbb{H}^n$

Hyperbolic space can be represented by several isometric models. We focus on the Upper Half-Plane model ($\mathbb{U}^n$), the Poincaré Ball model ($\mathbb{B}^n$), and the Hyperboloid model.

Metrics and Transformations

- **Upper Half-Plane Model** $\mathbb{U}^n$: Defined on $\{(x_1, \ldots, y) \in \mathbb{R}^n : y > 0\}$. In dimension 2 ($z = x + iy$):
$$g_{\mathbb{U}} = \frac{dx^2 + dy^2}{y^2} = \frac{dz d\bar{z}}{(\operatorname{Im} z)^2}$$
- **Poincaré Ball Model** $\mathbb{B}^n$: Defined on $\{w \in \mathbb{R}^n : |w| < 1\}$. In dimension 2 ($w = u + iv$):
$$g_{\mathbb{B}} = 4\frac{du^2 + dv^2}{(1 - |w|^2)^2} = 4\frac{dw d\bar{w}}{(1 - |w|^2)^2}$$

Cayley Transform: The isometry $X : \mathbb{B}^2 \to \mathbb{U}^2$ is given by the Möbius transformation:
$$z = X(w) = -i\frac{w + i}{w - i}$$

Verification of Isometry. 1. Compute the differential dz:
$$dz = -i\frac{(w - i) - (w + i)}{(w - i)^2}dw = -i\frac{-2i}{(w - i)^2}dw = \frac{-2}{(w - i)^2}dw$$
Thus, $dz d\bar{z} = \frac{4}{|w-i|^4} dw d\bar{w}$.

2. Compute the imaginary part $\operatorname{Im} z$:
$$\operatorname{Im} z = \frac{1}{2i}(z - \bar{z}) = \frac{1}{2i}\left(-i\frac{w + i}{w - i} - i\frac{\bar{w} - i}{\bar{w} + i}\right) = -\frac{1}{2}\left(\frac{(w + i)(\bar{w} + i) + (\bar{w} - i)(w - i)}{|w - i|^2}\right)$$
$$\operatorname{Im} z = -\frac{1}{2}\left(\frac{(w\bar{w} + iw + i\bar{w} - 1) + (w\bar{w} - iw - i\bar{w} - 1)}{|w - i|^2}\right) = \frac{1 - |w|^2}{|w - i|^2}$$

3. Substitute into the metric $g_{\mathbb{U}}$:
$$\frac{dz d\bar{z}}{(\operatorname{Im} z)^2} = \frac{4/|w - i|^4}{(1 - |w|^2)^2/|w - i|^4} dw d\bar{w} = \frac{4 dw d\bar{w}}{(1 - |w|^2)^2} = g_{\mathbb{B}}$$

□

5.2 Isometry Groups

The groups of orientation-preserving isometries for these models are isomorphic:

- $\operatorname{Isom}^+(\mathbb{U}^2) \cong PSL(2, \mathbb{R}) = \{\begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R}, ad - bc = 1\}/\{\pm I\}$.
- $\operatorname{Isom}^+(\mathbb{B}^2) \cong PSU(1, 1) = \{\begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1\}/\{\pm I\}$.
- $\operatorname{Isom}^+(\mathbb{H}^2) \cong SO^+(2, 1)$.

Hence, $PSL(2, \mathbb{R}) \cong PSU(1, 1) \cong SO^+(2, 1)$.

5.3 Geometric Schwarz Lemma

Schwarz Lemma (Classical): If $f : \mathbb{B}^2 \to \mathbb{B}^2$ is holomorphic and $f(0) = 0$, then $|f'(0)| \leq 1$.

Theorem 5.1 (Geometric Schwarz Lemma)**.** *If $f : \mathbb{B}^2 \to \mathbb{B}^2$ is a holomorphic map, then for all $w_1, w_2 \in \mathbb{B}^2$:*

$$d_{hyp}(f(w_1), f(w_2)) \leq d_{hyp}(w_1, w_2)$$

where d_{hyp} is the distance induced by the hyperbolic metric.

Proof. **Step 1:** Show that for any $X \in T_p\mathbb{B}^2$, $|df_p(X)|_{\text{hyp}} \leq |X|_{\text{hyp}}$. Since $\text{Isom}^+(\mathbb{B}^2)$ acts transitively, we can assume $p = 0$ and $f(p) = 0$ by pre- and post-composing with isometries. At the origin, the hyperbolic metric is related to the Euclidean metric by $|X|_{\text{hyp}} = 2|X|_{\text{euc}}$. By the classical Schwarz Lemma, $|f'(0)| \leq 1$. Thus:

$$|df_0(X)|_{\text{hyp}} = 2|f'(0)X|_{\text{euc}} = |f'(0)| \cdot 2|X|_{\text{euc}} \leq |X|_{\text{hyp}}$$

Step 2: For any curve γ, $L(f \circ \gamma) \leq L(\gamma)$. Since the distance is the infimum of lengths of admissible curves, the inequality $d_{\text{hyp}}(f(w_1), f(w_2)) \leq d_{\text{hyp}}(w_1, w_2)$ follows. □

5.4 Connections on Vector Bundles

Directional Derivative on Functions

For $F \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, $D_X F = XF$ satisfies: 1. $\mathbb{R}$-linearity in X and F. 2. C^∞-linearity in X: $D_{fX}F = fD_X F$. 3. Leibniz rule: $D_X(fF) = (Xf)F + f(XF)$.

The Connection ∇

Let $E \to M$ be a real vector bundle. A **connection** on E is a map $\nabla : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$, denoted $(X, \xi) \mapsto \nabla_X \xi$, satisfying:

1. $\nabla_X \xi$ is $\mathbb{R}$-linear in X and ξ.
2. $C^\infty(M)$**-linearity in** X: $\nabla_{fX}\xi = f\nabla_X \xi$.
3. **Leibniz Rule in** ξ: $\nabla_X(f\xi) = (Xf)\xi + f\nabla_X \xi$.

Proposition 5.2. *$\nabla_X \xi(p)$ depends only on the value of X at p and the values of ξ in an arbitrarily small neighborhood of p.*

Proof. In a coordinate neighborhood U with $X = \sum X^i \partial_i$, we have $\nabla_X \xi = \sum X^i \nabla_{\partial_i} \xi$. If $X(p) = 0$, then $X^i(p) = 0$, so $\nabla_X \xi(p) = 0$. This shows point-wise dependence on X. For ξ, if $\xi \equiv \tilde{\xi}$ on U, then $\nabla_X(\xi - \tilde{\xi}) = 0$ at p by the Leibniz rule. □

Curvature Tensor

Definition 5.3. The curvature $F(X, Y)\xi$ (often denoted $R(X, Y)\xi$) is defined as:

$$F(X, Y)\xi = \nabla_X \nabla_Y \xi - \nabla_Y \nabla_X \xi - \nabla_{[X,Y]}\xi$$

It can be shown that F is $C^\infty(M)$-linear in X, Y, and ξ, hence it is a tensor field.

6 Lecture 6

6.1 The Curvature Tensor

Definition 6.1. Let ∇ be a connection on a vector bundle $E \to M$. The **curvature operator** $F(X,Y) : \Gamma(E) \to \Gamma(E)$ is defined by:

$$F(X,Y)\xi = \nabla_X \nabla_Y \xi - \nabla_Y \nabla_X \xi - \nabla_{[X,Y]}\xi$$

for vector fields $X, Y \in \mathfrak{X}(M)$ and a section $\xi \in \Gamma(E)$.

Proposition 6.2. *F is $C^\infty(M)$-multilinear in X, Y, ξ.*

Proof for linearity in ξ. Let $f \in C^\infty(M)$. We want to show $F(X,Y)(f\xi) = fF(X,Y)\xi$.

$$\begin{aligned}
F(X,Y)(f\xi) &= \nabla_X \nabla_Y (f\xi) - \nabla_Y \nabla_X (f\xi) - \nabla_{[X,Y]}(f\xi) \\
&= \nabla_X \left((Yf)\xi + f\nabla_Y \xi\right) - \nabla_Y \left((Xf)\xi + f\nabla_X \xi\right) - \left(([X,Y]f)\xi + f\nabla_{[X,Y]}\xi\right) \\
&= (X(Yf)\xi + (Yf)\nabla_X \xi + (Xf)\nabla_Y \xi + f\nabla_X \nabla_Y \xi) \\
&\quad - (Y(Xf)\xi + (Xf)\nabla_Y \xi + (Yf)\nabla_X \xi + f\nabla_Y \nabla_X \xi) \\
&\quad - \left((X(Yf) - Y(Xf))\xi + f\nabla_{[X,Y]}\xi\right) \\
&= f\left(\nabla_X \nabla_Y \xi - \nabla_Y \nabla_X \xi - \nabla_{[X,Y]}\xi\right) = fF(X,Y)\xi
\end{aligned}$$

The terms involving derivatives of f cancel out completely. □

Corollary 6.3. *F is a section of the bundle $\Lambda^2 M \otimes End(E)$. If $E = TM$, we call ∇ a **linear connection**.*

6.2 Linear Connections in Local Coordinates

Let $\{E_i\}$ be a local frame for TM (typically coordinate vectors ∂_i). We define the **Christoffel symbols** Γ_{ij}^k by:

$$\nabla_{E_i} E_j = \Gamma_{ij}^k E_k$$

For vector fields $X = X^i E_i$ and $Y = Y^j E_j$, the covariant derivative is:

$$\nabla_X Y = \sum_k \left(X(Y^k) + \sum_{i,j} \Gamma_{ij}^k X^i Y^j \right) E_k$$

Notation: In coordinate basis ∂_i, we often denote:

$$\nabla_{\partial_i}(V^j \partial_j) = V^j_{;i} \partial_j, \quad \text{where } V^j_{;i} = \partial_i V^j + \Gamma_{ik}^j V^k$$

6.3 Induced Connections

Connection on the Dual Bundle T^*M

We wish to define a connection on T^*M such that the Leibniz rule holds for the pairing $\omega(Y)$:

$$X(\omega(Y)) = (\nabla_X \omega)(Y) + \omega(\nabla_X Y)$$

Definition 6.4. The induced connection on T^*M is given by:

$$(\nabla_X \omega)(Y) = X(\omega(Y)) - \omega(\nabla_X Y)$$

Proposition 6.5. *This defines a valid connection on T^*M.*

Verification of 1-form property. For $f \in C^\infty(M)$, we must show $(\nabla_X \omega)(fY) = f(\nabla_X \omega)(Y)$:

$$\begin{aligned}(\nabla_X \omega)(fY) &= X(\omega(fY)) - \omega(\nabla_X(fY)) \\ &= X(f\omega(Y)) - \omega((Xf)Y + f\nabla_X Y) \\ &= (Xf)\omega(Y) + fX(\omega(Y)) - (Xf)\omega(Y) - f\omega(\nabla_X Y) \\ &= f\left(X(\omega(Y)) - \omega(\nabla_X Y)\right) = f(\nabla_X \omega)(Y)\end{aligned}$$

Similar proofs hold for C^∞-linearity in X and the Leibniz rule in ω. □

Local Coordinate Form: For $\omega = \omega_j dx^j$, the covariant derivative is:

$$\nabla_{\partial_i}(\omega_j dx^j) = \omega_{j;i} dx^j, \quad \text{where } \omega_{j;i} = \partial_i \omega_j - \Gamma^k_{ij}\omega_k$$

Connection on Tensor Products

A connection can be induced on the tensor bundle $TM \otimes TM$:

$$\nabla_X(Y \otimes Z) = (\nabla_X Y) \otimes Z + Y \otimes (\nabla_X Z)$$

Covariant Derivative Along a Curve

Setup: Let $\gamma : I \to M$ be a smooth curve.

- Velocity vector field: $\dot{\gamma}(t) = d\gamma_t(\partial_t)$.
- Vector field along γ: A map $V : I \to TM$ such that $V(t) \in T_{\gamma(t)}M$. This is a section of the pullback bundle $\gamma^* TM$.
- **Extendability:** $V \in \mathfrak{X}(\gamma)$ is extendable if there exists a vector field $\tilde{V}$ in a neighborhood of γ such that $\tilde{V}_{\gamma(t)} = V(t)$.

Proposition: Given a connection ∇ on M, there is a unique operator $D_t : \mathfrak{X}(\gamma) \to \mathfrak{X}(\gamma)$ satisfying:

1. D_t is $\mathbb{R}$-linear.
2. Leibniz Rule: $D_t(fV) = \frac{df}{dt}V + fD_tV$ for $f \in C^\infty(I)$.
3. Consistency: If V extends to $\tilde{V}$, then $(D_tV)(t) = \nabla_{\dot{\gamma}(t)}\tilde{V}$.

Local Coordinate Form: If $V(t) = V^i(t)\partial_i$, then:

$$D_tV = \sum_k \left(\frac{dV^k}{dt} + \sum_{i,j} \Gamma^k_{ij} \frac{d\gamma^i}{dt} V^j \right) \partial_k$$

7 Lecture 7

7.1 Geodesics

Definition 7.1. A smooth curve $\gamma : I \to M$ is a **geodesic** with respect to a connection ∇ if its velocity vector field is parallel along the curve, i.e.,

$$D_t\dot{\gamma} = 0.$$

Local Coordinates: In a local coordinate system $\{x^i\}$, let $\gamma(t) = (\gamma^1(t), \ldots, \gamma^n(t))$. The velocity is $\dot{\gamma} = \frac{d\gamma^i}{dt}\partial_i$. The **geodesic equation** is a system of second-order non-linear ODEs:

$$\frac{d^2\gamma^k}{dt^2} + \sum_{i,j} \Gamma^k_{ij}(\gamma(t))\frac{d\gamma^i}{dt}\frac{d\gamma^j}{dt} = 0, \quad \text{for } k = 1, \ldots, n.$$

Existence and Uniqueness: By standard ODE theory (Picard-Lindelöf), given $p \in M$ and $v \in T_pM$, there exists a unique maximal geodesic $\gamma : I \to M$ such that $\gamma(0) = p$ and $\dot{\gamma}(0) = v$.

- **Example:** In $M = \mathbb{R}^2 \setminus \{0\}$, a geodesic starting at $(-1, 0)$ with velocity $(1, 0)$ is defined only for $t \in (-\infty, 1)$.

7.2 Parallel Transport

Definition 7.2. A vector field V along γ is **parallel** if $D_tV = 0$.

Parallel Transport Map: For a curve $\gamma : [t_0, t_1] \to M$, the map $P_{t_0,t_1} : T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$ defined by $v_0 \mapsto V(t_1)$ (where V is parallel and $V(t_0) = v_0$) is a linear isomorphism.

Relationship to Covariant Derivative: The covariant derivative can be recovered from parallel transport:

$$(D_tV)(t_0) = \lim_{t\to t_0} \frac{P_{t,t_0}V(t) - V(t_0)}{t - t_0} = \left.\frac{d}{dt}\right|_{t=t_0} P_{t,t_0}V(t).$$

Proof Sketch: Use a parallel frame $\{E_i(t)\}$ such that $D_tE_i = 0$. Write $V(t) = v^i(t)E_i(t)$, then $P_{t,t_0}V(t) = v^i(t)E_i(t_0)$. Taking the derivative yields the component-wise derivative.

7.3 Connections on Submanifolds of $\mathbb{R}^n$

Let $M \subset \mathbb{R}^n$ be an embedded submanifold. Let $\Pi^\top : T_p\mathbb{R}^n \to T_pM$ denote the orthogonal projection onto the tangent space.

Induced Connection: Let $\bar{\nabla}$ be the standard Euclidean connection on $\mathbb{R}^n$. For vector fields X, Y on M, define:

$$\nabla_X Y = \Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y})$$

where $\bar{X}, \bar{Y}$ are local extensions of X, Y to $\mathbb{R}^n$.

Properties:

1. **Metric Compatibility:** $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$.
2. **Symmetry (Torsion-free):** $\nabla_X Y - \nabla_Y X = [X, Y]$.

These properties confirm that the induced connection is the **Levi-Civita connection** for the induced Riemannian metric on M.

Proof of Connection Properties on Submanifolds

Let $M \subset \mathbb{R}^n$ be an embedded submanifold. Let $\bar{\nabla}$ denote the standard Euclidean connection on $\mathbb{R}^n$, and $\Pi^\top : T_p\mathbb{R}^n \to T_pM$ be the orthogonal projection onto the tangent space. The induced connection on M is defined as $\nabla_X Y = \Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y})$, where $\bar{X}, \bar{Y}$ are local extensions of X, Y.

Metric Compatibility

Property: $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$.

Proof. The standard connection $\bar{\nabla}$ on $\mathbb{R}^n$ is compatible with the Euclidean metric, satisfying:

$$\bar{X}\langle \bar{Y}, \bar{Z}\rangle = \langle \bar{\nabla}_{\bar{X}}\bar{Y}, \bar{Z}\rangle + \langle \bar{Y}, \bar{\nabla}_{\bar{X}}\bar{Z}\rangle$$

When restricted to the manifold M, the term $\bar{X}\langle \bar{Y}, \bar{Z}\rangle$ becomes $X\langle Y, Z\rangle$. Since $Z \in T_pM$ is a tangent vector, the inner product only "sees" the tangential component of any vector it is paired with. Thus:

$$\langle \bar{\nabla}_{\bar{X}}\bar{Y}, Z\rangle = \langle \Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y}), Z\rangle = \langle \nabla_X Y, Z\rangle$$

Similarly, $\langle Y, \bar{\nabla}_{\bar{X}}\bar{Z}\rangle = \langle Y, \nabla_X Z\rangle$. Substituting these back into the Euclidean identity yields:

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$

□

Symmetry (Torsion-free)

Property: $\nabla_X Y - \nabla_Y X = [X, Y]$.

Proof. The standard connection $\bar{\nabla}$ on $\mathbb{R}^n$ is torsion-free, meaning:

$$\bar{\nabla}_{\bar{X}}\bar{Y} - \bar{\nabla}_{\bar{Y}}\bar{X} = [\bar{X}, \bar{Y}]$$

Applying the orthogonal projection $\Pi^\top$ to both sides gives:

$$\Pi^\top(\bar{\nabla}_{\bar{X}}\bar{Y}) - \Pi^\top(\bar{\nabla}_{\bar{Y}}\bar{X}) = \Pi^\top([\bar{X}, \bar{Y}])$$

By the definition of the induced connection, the left-hand side is $\nabla_X Y - \nabla_Y X$. Furthermore, since M is a smooth submanifold, the Lie bracket of two tangent vector fields X and Y is itself tangent to M, implying $\Pi^\top([X, Y]) = [X, Y]$. Therefore:

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

□

8 Lecture 8

8.1 Holonomy Group

Definition 8.1. Let ∇ be a linear connection on M. Given a loop (curve) $\gamma : [t_0, t_1] \to M$ at $p = \gamma(t_0) = \gamma(t_1)$, the parallel transport $P_\gamma : T_pM \to T_pM$ is a linear automorphism. We note $H(\gamma) \subset Aut(T_pM)$ be the automorphisms induced by the loop γ. The **Holonomy group** at p is defined as:

$$\mathrm{Hol}_p(\nabla) = \{P_\gamma \in H(\gamma) \mid \gamma \text{ is a loop at } p\}.$$

Property: If p, p' are in the same connected component of M, then $\mathrm{Hol}_p(\nabla) \cong \mathrm{Hol}_{p'}(\nabla)$ via conjugation by parallel transport along a path connecting p and p'.

Example 8.2. In $\mathbb{R}^n$ with the standard connection, $\mathrm{Hol}_p(\nabla) = \{I\}$ for all p since parallel transport along any loop is the identity.

8.2 Metric Compatibility

Definition 8.3. A connection ∇ is **metric-compatible** if:

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle, \quad \forall X, Y, Z \in \mathfrak{X}(M).$$

Proposition 8.4. ***TFAE:***

1. *∇ is metric-compatible.*
2. *$\nabla g = 0$.*
3. *$\frac{d}{dt}\langle V, W\rangle = \langle D_t V, W\rangle + \langle V, D_t W\rangle$ for vector fields along any curve γ.*
4. *Parallel transport $P_{t_0,t_1} : T_{\gamma(t_0)}M \to T_{\gamma(t_1)}M$ is an isometry.*
5. *$Hol_p(\nabla) \subset O(n)$ (after choosing an orthonormal basis).*

Proof. We follow the logic from the lecture notes to prove the equivalences.

$1 \iff 2$

Using the formula for the covariant derivative of a $(0, 2)$-tensor g:

$$X(g(Y, Z)) = (\nabla_X g)(Y, Z) + g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$$

Since $g(Y, Z) = \langle Y, Z\rangle$, $\nabla g = 0$ is clearly equivalent to the metric compatibility condition.

$1 \implies 3$

Define the operator $S(V, W) = \frac{d}{dt}\langle V, W\rangle - \langle D_t V, W\rangle - \langle V, D_t W\rangle$ for $V, W \in \mathfrak{X}(\gamma)$.

1. S is $C^\infty(I)$-linear. For $f \in C^\infty(I)$, the term f' from $\frac{d}{dt}(f\langle V, W\rangle)$ cancels with the f' from $\langle D_t(fV), W\rangle$.
2. Since S is linear, it suffices to check it on a local frame. For any t_0, extend $V(t_0), W(t_0)$ to vector fields Y, Z on M and let $X = \gamma'(t_0)$.
3. Then $S(V, W)(t_0) = X\langle Y, Z\rangle - \langle \nabla_X Y, Z\rangle - \langle Y, \nabla_X Z\rangle$, which is zero by condition 1. Thus $S \equiv 0$.

$3 \implies 4$

If V_1, V_2 are parallel, then $D_t V_1 = 0$ and $D_t V_2 = 0$. Substituting into condition 3: $\frac{d}{dt}\langle V_1, V_2\rangle = \langle 0, V_2\rangle + \langle V_1, 0\rangle = 0$. Thus the inner product is constant.

$4 \implies 3$

Let $\{E_i(t)\}$ be an orthonormal basis at $\gamma(t_0)$ extended by parallel transport along γ. By condition 4, $\{E_i(t)\}$ remains an orthonormal basis for all t (since $D_t E_i = 0$). Let $V(t) = V^i(t)E_i(t)$ and $W(t) = W^j(t)E_j(t)$. Then $\langle V, W\rangle = \sum V^i W^i$. Taking the derivative: $\frac{d}{dt}\langle V, W\rangle = \sum(\dot{V}^i W^i + V^i \dot{W}^i)$. Meanwhile, $D_t V = \dot{V}^i E_i$ implies $\langle D_t V, W\rangle = \sum \dot{V}^i W^i$. The identity holds.

$4 \iff 5$

By definition, $P_{t_0,t_1} V(t_0) = V(t_1)$ where V is a parallel vector field. Condition 4 states $\langle V(t_0), \tilde{V}(t_0)\rangle = \langle V(t_1), \tilde{V}(t_1)\rangle$, which means P_{t_0,t_1} preserves the inner product, hence it is an isometry. $\square$

8.3 Torsion and Levi-Civita Connection

Definition 8.5. The **Torsion tensor** T is defined by:

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

A connection is **torsion-free** if $T = 0$.

Theorem 8.6 (Fundamental Theorem of Riemannian Geometry)**.** *On any Riemannian manifold (M, g), there exists a unique connection ∇ that is both metric-compatible and torsion-free. This is the **Levi-Civita connection**.*

The one and only connection is given by:
Koszul Formula:

$$\begin{aligned} 2\langle \nabla_X Y, Z\rangle = {} & X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle \\ & + \langle [X, Y], Z\rangle - \langle [Y, Z], X\rangle + \langle [Z, X], Y\rangle. \end{aligned}$$

8.4 Invariant Metrics and Compact Groups

If the holonomy group $K = \mathrm{Hol}_p(\nabla)$ is a compact subgroup of $\mathrm{Aut}(T_p M)$, we can construct an invariant inner product by averaging over the Haar measure μ of K:

$$\widetilde{\langle v, w\rangle} = \int_K \langle kv, kw\rangle_0 \, d\mu(k).$$

This implies that a connection comes from some Riemannian metric if and only if its holonomy group is contained in a conjugate of $O(n)$.

9 Lecture 9

9.1 Holonomy Groups

Definition 9.1 (Restricted Holonomy Group)**.** The restricted holonomy group $\mathrm{Hol}_0(p)$ consists of parallel translations around contractible loops based at p.

fact:$\mathrm{Hol}_0(p)$ is the identity component of $\mathrm{Hol}(p)$. It is a connected, closed Lie subgroup of $SO(n)$.

Definition 9.2 (Special Holonomy)**.** Generally, $\mathrm{Hol}(p) = O(n)$. If $\mathrm{Hol}(p) \subsetneq O(n)$, the manifold is said to have *special holonomy* (e.g., Kähler, G2, etc.).

9.2 Levi-Civita Connection and Christoffel Symbols

Theorem 9.3 (Koszul Formula)**.** *The unique Levi-Civita connection is determined by:*

$$2\langle \nabla_X Y, Z\rangle = X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle + \langle [X, Y], Z\rangle - \langle [Y, Z], X\rangle + \langle [Z, X], Y\rangle$$

Remark 9.4 (In Local Coordinates). Let ∂_i be coordinate vectors. Then $\langle \nabla_{\partial_i}\partial_j, \partial_l\rangle = \frac{1}{2}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$. We define the Christoffel symbols of the first and second kind:

$$\Gamma_{ijl} = \langle \nabla_{\partial_i}\partial_j, \partial_l\rangle, \quad \nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k \implies \Gamma^k_{ij} = g^{kl}\Gamma_{ijl}$$

Remark 9.5 (Symmetry). Since $\nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i = [\partial_i, \partial_j] = 0$, we have $\Gamma^k_{ij} = \Gamma^k_{ji}$.

9.3 Properties of Geodesics

Proposition 9.6. *If ∇ is the Levi-Civita connection and γ is a geodesic, then $|\dot\gamma|$ is constant.*

Proof. $\frac{d}{dt}|\dot\gamma|^2 = \frac{d}{dt}\langle\dot\gamma, \dot\gamma\rangle = 2\langle D_t\dot\gamma, \dot\gamma\rangle$. Since γ is a geodesic, $D_t\dot\gamma = 0$, hence the derivative is zero. □

9.4 Isometries and Geodesics

Proposition 9.7. *If $\phi : M \to \tilde M$ is an isometry, then:*

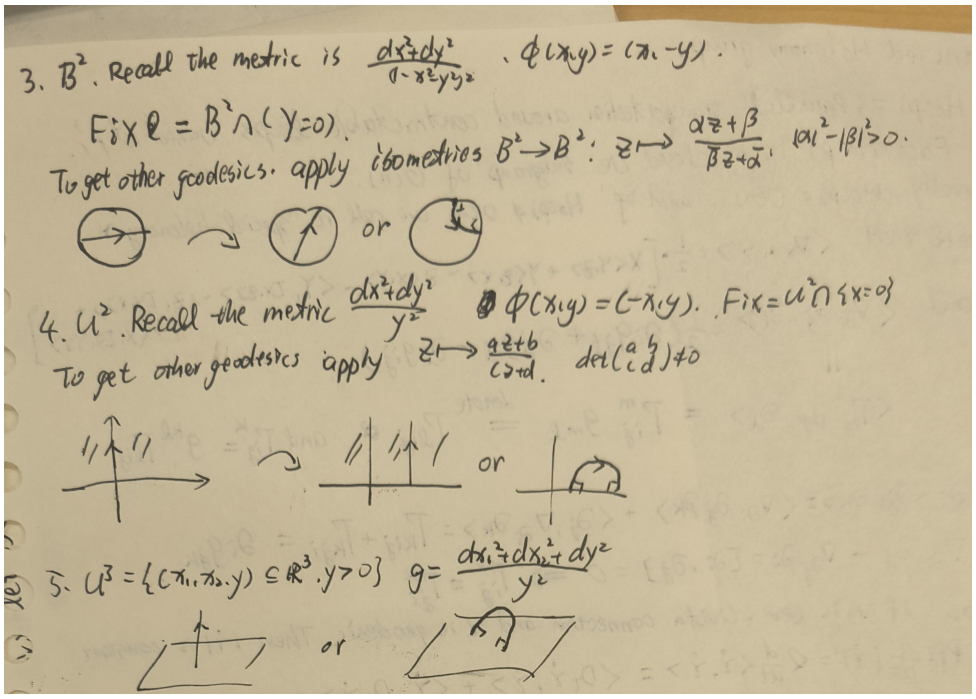
1. *$\phi_*(\nabla_X Y) = \tilde\nabla_{\phi_* X}\phi_* Y$. (Isometries preserve the Levi-Civita connection).*
2. *γ is a geodesic on $M \iff \phi\circ\gamma$ is a geodesic on $\tilde M$.*

Proof. **(1)** First, define $\phi_*\tilde\nabla$ by $(\phi_*\tilde\nabla)_X Y = \phi_*^{-1}(\tilde\nabla_{\phi_* X}\phi_* Y)$, and proof that it is a metric-compatible, torsion-free connection. Hence by unuqueness of Levi-Civita connection, $\phi_*\tilde\nabla = \nabla$.
(2) Trivial. □

Proposition 9.8 (Symmetry and Geodesics)**.** *If $\phi \in Isom(M)$ satisfies $\phi(p) = p$ and $d\phi_p(X) = X$, then the maximal geodesic γ with $\gamma(0) = p, \dot\gamma(0) = X$ is pointwise fixed by ϕ, i.e., $\phi(\gamma(t)) = \gamma(t)$.*

Example 9.9 (S^n)**.** Reflection through a 2-plane P spanned by p, X. The fixed point set $P\cap S^n$ is a great circle (geodesic).

Example 9.10 ($\mathbb{H}^n, \mathbb{B}^2, \mathbb{U}^2$)**.** Geodesics are obtained as fixed point sets of reflections (lines or circles orthogonal to the boundary).



9.5 The Exponential Map

Lemma 9.11 (Scaling)**.** *For a geodesic $\gamma_v(t)$, we have $\gamma_{cv}(t) = \gamma_v(ct)$.*

Definition 9.12 (Exponential Map)**.** Let $\mathcal{E} = \{v \in TM \mid \gamma_v(t) \text{ is defined for } t \in [0,1]\}$. The **exponential map** $\exp : \mathcal{E} \to M$ is defined by:

$$\exp(v) = \gamma_v(1), \quad \exp_p(v) = \exp|_{T_pM}(v)$$

Proposition 9.13 (Properties of $\exp_p$)**.**

1. *$\mathcal{E}$ is an open subset of TM containing the zero section.*
2. $\exp_p(tv) = \gamma_v(t)$.
3. $\exp_p$ *is smooth.*
4. $d(\exp_p)_0 : T_0(T_pM) \cong T_pM \to T_pM$ *is the identity map.*

Remark 9.14 (Motivation: Geodesic Flow). The geodesic equation $\ddot{x}^k + \Gamma^k_{ij}\dot{x}^i\dot{x}^j = 0$ can be written as a first-order system on TM:

$$\begin{cases} \frac{dx^i}{dt} = v^i \\ \frac{dv^k}{dt} = -\Gamma^k_{ij}v^iv^j \end{cases}$$

This defines a vector field G on TM whose flow gives the geodesics.

Proof of Properties of $\exp_p$. **$\mathcal{E}$ is an Open Subset of TM and $\exp$ is Smooth:**
Logic from Blackboard 2: Let $\Theta_t : \mathcal{D} \to TM$ be the flow generated by the vector field G. According to the theory of Ordinary Differential Equations (ODEs), the flow $\Theta_t(v)$ is defined on an open domain $\mathcal{D} \subset \mathbb{R} \times TM$.

We define the domain of the exponential map as:

$$\mathcal{E} = \{v \in TM \mid \Theta_t(v) \text{ is defined for } t \in [0,1]\}$$

If the flow starting from v exists up to time 1, then by the **fundamental theorem of ODEs** (continuous dependence on initial conditions), the flow exists up to time 1 (and "a bit farther") for all vectors in a sufficiently small neighborhood of v in TM. Hence, $\mathcal{E}$ is an **open subset** of TM.

Furthermore, since the geodesic vector field G is smooth, its flow Θ_t is also smooth. The exponential map is defined as the composition:

$$\exp = \pi \circ \Theta_1$$

where $\pi : TM \to M$ is the natural projection $(x, v) \mapsto x$. As a composition of smooth maps, $\exp : \mathcal{E} \to M$ is **smooth**.

$\exp_p(tv) = \gamma_v(t)$:
By the definition of the exponential map, $\exp_p(v) = \gamma_v(1)$. Recall the **scaling property** (homogeneity) of geodesics: for any constant $c > 0$, the curve $t \mapsto \gamma_v(ct)$ is a geodesic with initial velocity $c\dot{\gamma}_v(0) = cv$. By uniqueness of geodesics, we have:

$$\gamma_{cv}(t) = \gamma_v(ct)$$

Setting $c = t$ and evaluating at 1, we obtain:

$$\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t \cdot 1) = \gamma_v(t)$$

This shows that $\exp_p$ maps the line segment in T_pM through the origin to the corresponding geodesic on M.

The Differential $(d\exp_p)_0 = \mathbf{id}_{T_pM}$:
We wish to compute the derivative of $\exp_p : T_pM \to M$ at the origin $0 \in T_pM$. We identify $T_0(T_pM)$ with T_pM itself. For any $v \in T_pM$, consider the path $c(t) = tv$ in T_pM. Note that $c(0) = 0$ and $c'(0) = v$.

By the definition of the differential:

$$(d\exp_p)_0(v) = \left.\frac{d}{dt}\right|_{t=0} \exp_p(c(t)) = \left.\frac{d}{dt}\right|_{t=0} \exp_p(tv)$$

Using the property derived in the previous section ($\exp_p(tv) = \gamma_v(t)$):

$$(d\exp_p)_0(v) = \left.\frac{d}{dt}\right|_{t=0} \gamma_v(t) = \dot{\gamma}_v(0)$$

By the initial conditions of the geodesic γ_v, we have $\dot{\gamma}_v(0) = v$. Therefore:

$$(d\exp_p)_0(v) = v$$

which means $(d\exp_p)_0$ is the **identity map** on T_pM. $\square$

10 Lecture 10

10.1 Exponential Map and Normal Coordinates

Lemma 10.1. *For every $p \in M$, there exists a neighborhood V of $0 \in T_pM$ and a neighborhood $\mathcal{U}$ of p such that $\exp_p : V \to \mathcal{U}$ is a diffeomorphism.*

Definition 10.2 (Normal Neighborhood)**.** A neighborhood of p is called a **normal neighborhood** if it is the diffeomorphic image of a star-shaped neighborhood of $0 \in T_pM$ under $\exp_p$.

Definition 10.3 (Geodesic Ball)**.** If $\varepsilon > 0$ is small enough such that $\exp_p : B_\varepsilon(0) \to B_\varepsilon(p)$ is a diffeomorphism, $B_\varepsilon(p)$ is called a **geodesic ball**. Its boundary is the **geodesic sphere**.

Riemannian Normal Coordinates (RNC)

Let $\{E_i\}$ be an orthonormal basis for T_pM. The coordinate map is $(x^i) \mapsto \exp_p(\sum x^i E_i)$.

Proposition 10.4 (Properties of RNC)**.** *1. p has coordinates $(0, \dots, 0)$.*

2. Geodesics through p are radial lines: $x^i(t) = tv^i$.

3. At the origin: $g_{ij}(0) = \delta_{ij}$.

4. At the origin: $\Gamma^k_{ij}(0) = 0$.

Proof. 1 and 2 are automatic.

3. Consider the pull back metric $\exp_p^* g$, we aim to show that $\exp_p^* g(0) = \delta_{ij}$. For $V = V^i \frac{\partial}{\partial x^i} \in T_0T_pM$. Put $\gamma(t) = tV$, $\exp_p(\gamma(t)) = \gamma_V(t)$. Then $(d\exp_p)_0(V) = \dot\gamma_V(0) = V$, i.e.

$$\exp_p^* g_0(V, V) = \|V\|^2 \Longrightarrow \exp_p^* g_0 = \delta_{ij}.$$

4. Let $\gamma(t) = tv$ be a geodesic. The geodesic equation is $\ddot x^i + \Gamma^i_{jk}\dot x^j \dot x^k = 0$. Substituting $x^i(t) = tv^i$, we have $0 + \Gamma^i_{jk}(0)v^j v^k = 0$. Since this holds for all $v \in T_pM$ and Γ^i_{jk} is symmetric, $\Gamma^i_{jk}(0) = 0$. □

10.2 Uniformly Normal Neighborhoods

Definition 10.5 (Uniformly Normal Neighborhood)**.** An open set $W \subset M$ is **uniformly normal** if there exists $\delta > 0$ such that for any $w \in W$, $W \subset \exp_w(B_\delta(0))$, and $\exp_w$ is a diffeomorphism on $B_\delta(0)$.

Proposition 10.6. *Every $p \in M$ has a uniformly normal neighborhood.*

Proof. Define the map $F : \mathcal{E} \to M \times M$ by $F(q, v) = (q, \exp_q v)$, where $\mathcal{E} \subset TM$ is the open domain where the exponential map is defined. We want to show that $dF_{(p,0)}$ is an isomorphism.

Recall that the tangent space $T_{(p,0)}TM$ fits into a short exact sequence involving the vertical space (tangent to the fiber T_pM) and the horizontal space (projecting to T_pM via $d\pi$). The tangent space to the product manifold is $T_{(p,p)}(M \times M) \cong T_pM \oplus T_pM$.

We can construct the following diagram:

$$\begin{tikzcd}
0 \arrow[r] & T_pM \arrow[r, "i_*"] \arrow[d, "\mathrm{id}"] & T_{(p,0)}TM \arrow[r, "d\pi"] \arrow[d, "dF_{(p,0)}"] & T_pM \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & T_pM \arrow[r, "0\oplus\mathrm{id}"] & T_{(p,p)}(M \times M) \arrow[r, "d\mathrm{pr}_1"] & T_pM \arrow[r] & 0
\end{tikzcd}$$

Let us verify that this diagram commutes:

- **Left Square:** For any $v \in T_pM$, the vertical lift $i_*(v)$ is the tangent vector to the curve $c(t) = (p, tv)$ in TM at $t = 0$. Applying the differential dF, we have:

$$dF_{(p,0)}(i_*(v)) = \frac{d}{dt}\bigg|_{t=0} F(p, tv) = \frac{d}{dt}\bigg|_{t=0} (p, \exp_p(tv)) = (0, v)$$

  which exactly matches the map $0 \oplus \mathrm{id}$ in the bottom row. Hence, the left square commutes.

- **Right Square:** Notice that for any $(q, v) \in TM$, projecting the first component gives $\mathrm{pr}_1(F(q, v)) = \mathrm{pr}_1(q, \exp_q v) = q = \pi(q, v)$. Taking the differential of both sides at $(p, 0)$ yields $d\mathrm{pr}_1 \circ dF_{(p,0)} = d\pi$, meaning the right square commutes.

Since the rows are exact and the two outer vertical maps are isomorphisms (they are identity maps), the **Five Lemma** implies that the middle map $dF_{(p,0)} : T_{(p,0)}TM \to T_{(p,p)}(M \times M)$ is a linear isomorphism.

By the **Inverse Function Theorem**, F is a local diffeomorphism at $(p, 0)$. Thus, there exists a neighborhood $V \subset TM$ of $(p, 0)$ that is mapped diffeomorphically onto a neighborhood of (p, p) in $M \times M$.

We can choose a sufficiently small open neighborhood $W \subset M$ of p and a radius $\delta > 0$ such that $W \times W \subset F(V \cap \{|v| < \delta\})$. Then, for any pair $(w, w') \in W \times W$, there exists a unique tangent vector $v \in T_wM$ with $|v| < \delta$ such that $F(w, v) = (w, w')$. By definition of F, this means $\exp_w(v) = w'$.

Furthermore, because F is a diffeomorphism on this restricted domain, for each fixed $w \in W$, the map $\exp_w$ acts as a diffeomorphism from the δ-ball $B_\delta(0) \subset T_wM$ onto its image, which completely contains W. Therefore, W is uniformly normal. □

10.3 Geodesic as length-minimizing curves

Remark 10.7 (Recall). Regular curve $\dot{\gamma} \neq 0$, admissible curve: piecewise regular curve.

Definition 10.8. Given intervals $I, J \in \mathbb{R}$. A 1-parameter family of curves is a continuous map $\Gamma : J \times I \to M$. We write $\Gamma_s(t) = \Gamma(s, t)$ be a curve for each s.

Definition 10.9. A vector field along Γ is a continuous map $V : J \times I \to TM$ s.t. $\pi \circ V = \Gamma$.

Definition 10.10. A 1-parameter family of curves is admissible if:

1. J is open and $I = [a, b]$.
2. $\exists$ partition $a = a_0 < a_1 < \cdots < a_k = b$, so Γ is smooth on $J \times [a_{i-1}, a_i]$.
3. $\Gamma_s(t)$ is admissible for each s.

Remark 10.11. $\partial_t\Gamma$ may not be a vector field along Γ, but $\partial_s\Gamma$ is.

Definition 10.12. Given $\gamma : [a, b] \to M$, a variation of γ is a 1-parameter family of curves (admissible), and $\Gamma(0, t) = \gamma(t)$.

Definition 10.13. If Γ is a variation of γ, then the variation field of γ is the piecewise smooth vector field

$$V(t) = \frac{\partial \Gamma}{\partial s}(0, t) \quad \text{along } \gamma.$$

Definition 10.14. A vector field along γ is proper if $V(a) = V(b) = 0$.

Remark 10.15. If Γ is a proper variation of γ, then the variation field is a proper vector field along γ.

Lemma 10.16. *If γ is admissible, and V is a vector field along γ, then $\exists$ variation Γ of γ, so V is its variation field.*
If V is proper then we can make Γ be proper.

Lemma 10.17. *On rectangles $J \times [a_{i-1}, a_i]$,*

$$D_s \partial_t \Gamma = D_t \partial_s \Gamma.$$

Proof. Say Γ lives in a coordinate neighborhood. $\Gamma(s,t) = (x^1(s,t), \ldots, x^n(s,t))$.

$$\partial_t \Gamma = \sum \frac{\partial x^i}{\partial t} \partial_i, \quad \partial_s \Gamma = \sum \frac{\partial x^i}{\partial s} \partial_i$$

$$D_t V = \sum \left(\frac{dv^i}{dt} + \Gamma^i_{jk} \frac{dx^j}{dt} v^k \right) \partial_i$$

Hence,

$$D_s \partial_t \Gamma = \sum \left(\frac{\partial^2 x^i}{\partial s \partial t} + \Gamma^i_{jk} \frac{\partial x^j}{\partial s} \frac{\partial x^k}{\partial t} \right) \partial_i$$

which is symmetric in s, t. $\square$

11 Lecture 11

11.1 Isometries and Geodesics

Proposition 11.1. *M, M' Riemannian manifold, s.t.*

1. *$p \in M$.*
2. *$\sigma_1, \sigma_2 : M \to M'$ isometries.*
3. *$\sigma_1(p) = \sigma_2(p)$.*
4. *$(d\sigma_1)_p = (d\sigma_2)_p$.*
5. *Every point $q \in M$ can be connected to p by a geodesic.*

then $\sigma_1 = \sigma_2$.

Proof. Put $\alpha = \sigma_1^{-1} \circ \sigma_2 : M \to M$ isometry. $\alpha(p) = p$, $d\alpha_p = \mathrm{id} : T_pM \to T_pM$.
Pick $q \in M$. Let $\gamma : [0, L] \to M$ be the geodesic connecting p and q.
$\alpha \circ \gamma$ is another geodesic with

$$(\alpha \circ \gamma)(0) = p, \quad (\alpha \circ \gamma)'(0) = \gamma'(0)$$

so $\alpha \circ \gamma = \gamma$. Hence $\alpha(q) = q$.
i.e., α is id. □

$\to$ We can use this prop to proof that $\mathrm{Isom}(S^n) = O(n+1)$.

Pf: Let $\sigma \in \mathrm{Isom}(S^n)$. $p \in S^n$, $\{E_i\}$ is ortho basis of T_pS^n.
Get $\sigma(p) \in S^n$. $\{(d\sigma)_p(E_i)\}$ ortho basis for $T_{\sigma(p)}S^n$.
$\exists A \in O(n+1)$ so

$$A(\vec{p}, \vec{E}_1, \dots, \vec{E}_n) = (\vec{\sigma}(p), \overrightarrow{d\sigma_p E_1}, \dots),$$

which uniquely determines the isom A.

11.2 The First Variation Formula

Theorem 11.2 (First Variation Formula)**.** *Let $\Gamma(s,t)$ be a variation of a unit-speed curve $\gamma(t)$. Let $L(s) = L(\Gamma_s)$ be the length of the curve $t \mapsto \Gamma(s,t)$. Then the first variation of arc length is given by:*

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \langle V(b), \dot{\gamma}(b)\rangle - \langle V(a), \dot{\gamma}(a)\rangle - \int_a^b \langle V(t), D_t\dot{\gamma}(t)\rangle \, dt$$

where D_t is the covariant derivative along γ.

Proof. Let $T = \partial_t\Gamma$ and $S = \partial_s\Gamma$. Note that at $s = 0$, $T = \dot{\gamma}$ and $S = V$. Furthermore, since γ is unit-speed, $|\dot{\gamma}| = 1$. The length function is:

$$L(s) = \int_a^b \langle T, T\rangle^{1/2} \, dt$$

Taking the derivative with respect to s:

$$\frac{d}{ds}L(s) = \int_a^b \frac{\partial}{\partial s}\langle T, T\rangle^{1/2}\, dt = \int_a^b \frac{1}{2}\langle T, T\rangle^{-1/2} \cdot 2\langle D_s T, T\rangle\, dt$$

Here we use the fact that the connection is metric-compatible, so $\partial_s\langle T, T\rangle = 2\langle D_s T, T\rangle$. Evaluating at $s = 0$, since $\langle T, T\rangle^{1/2} = |\dot{\gamma}| = 1$, we have:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \int_a^b \langle D_s T, T\rangle\, dt$$

By the **Symmetry Lemma**, $D_s T = D_s(\partial_t \Gamma) = D_t(\partial_s \Gamma) = D_t S$. Substituting this in:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \int_a^b \langle D_t S, T\rangle\, dt$$

Using the product rule for the covariant derivative: $\frac{\partial}{\partial t}\langle S, T\rangle = \langle D_t S, T\rangle + \langle S, D_t T\rangle$, we can integrate by parts:

$$\begin{aligned}\int_a^b \langle D_t S, T\rangle\, dt &= \int_a^b \left(\frac{\partial}{\partial t}\langle S, T\rangle - \langle S, D_t T\rangle\right) dt \\ &= \Big[\langle S, T\rangle\Big]_a^b - \int_a^b \langle S, D_t T\rangle\, dt\end{aligned}$$

Evaluating at $s = 0$ where $S = V$, $T = \dot{\gamma}$, and $D_t T = D_t \dot{\gamma}$, we obtain the desired formula:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \langle V(b), \dot{\gamma}(b)\rangle - \langle V(a), \dot{\gamma}(a)\rangle - \int_a^b \langle V(t), D_t\dot{\gamma}\rangle\, dt$$

□

In the general case where γ is admissible, and $(a_1, ..., a_{k-1})$ are the break points, then its first variation formula is:

$$\left.\frac{d}{ds}\right|_{s=0} L(s) = \sum_{i=1}^{k-1}\langle V(a_i), \dot{\gamma}(a_i^+) - \dot{\gamma}(a_i^-)\rangle + \langle V(b), \dot{\gamma}(b)\rangle - \langle V(a), \dot{\gamma}(a)\rangle - \int_a^b \langle V(t), D_t\dot{\gamma}\rangle dt$$

We sometimes write $\dot{\gamma}(a_i^+) - \dot{\gamma}(a_i^-)$ as $\Delta_i\dot{\gamma}$.

Corollary 11.3. *Any length-minimizing curve between two points (parameterized by arc length) is a smooth geodesic.*

Proof. Let γ be a length-minimizing curve p and q. Fix t and let $\phi \in C^\infty([a, b])$ so that $\phi > 0$ on the smooth segment (a_{i-1}, a_i) for each i and $\phi = 0$ elsewhere.

Take $V(t) = \phi(t)D_t\dot{\gamma}(t)$ be a proper variation field, and $\Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$ be a proper variation of γ on (a_{i-1}, a_i). Then the first variation formula gives:

$$0 = \left.\frac{d}{ds}\right|_{s=0} L(\Gamma_s) = -\int_{a_{i-1}}^{a_i} \phi(t)\langle D_t\dot{\gamma}(t), D_t\dot{\gamma}(t)\rangle\, dt \leq 0$$

Hence, $D_t\dot{\gamma} = 0$, i.t. γ is a geodesic on each smooth segment.

Next, consider the whole curve γ. At each break point a_i, we construct a proper variation field V so that $V(a_i) = \Delta_i\dot{\gamma}$ and equals 0 outside a small neighborhood of a_i. Then the first variation formula gives:

$$0 = \left.\frac{d}{ds}\right|_{s=0} L(\Gamma_s) = -\langle V(a_i), \Delta_i\dot{\gamma}\rangle = -|\Delta_i\dot{\gamma}|^2 \leq 0$$

Hence, $\Delta_i\dot{\gamma} = 0$, i.e. the geodesic segments on either side of a_i match up smoothly. Therefore, γ is a smooth geodesic.

□

11.3 Geodesic Polar Coordinates and Gauss Lemma

Definition 11.4 (Geodesic Polar Coordinates)**.** Let $p \in M$. For a point in a normal neighborhood of p, we can assign polar coordinates (r, θ) where $r > 0$ and $\theta \in S^{n-1} \subset T_pM$, via the mapping:

$$(r, \theta) \mapsto \exp_p(r\theta)$$

The coordinate fields are ∂_r (radial direction) and ∂_{θ^i} (tangent to the geodesic sphere $S_r(p)$).

Proposition 11.5. *On a normal basis, $|\partial_r| = 1$. Where, $\partial_r = \frac{x^i}{r(x)} \frac{\partial}{\partial x^i}$*

Lemma 11.6 (Gauss Lemma)**.** *In a normal neighborhood of p, the radial geodesics are orthogonal to the geodesic spheres centered at p. That is, $\langle \partial_r, \partial_{\theta^i} \rangle = 0$.*

Proof. Let $\sigma : (-\epsilon, \epsilon) \to S^{n-1} \subset T_pM$ be a curve on the unit sphere in the tangent space. Define a variation by radial geodesics:

$$\Gamma(s, t) = \exp_p(t\sigma(s))$$

Here, $T = \partial_t \Gamma = \partial_r$ is the tangent vector to the radial geodesics, and $S = \partial_s \Gamma = \partial_\theta$ represents a vector field tangent to the geodesic spheres. Notice that for each fixed s, $t \mapsto \exp_p(t\sigma(s))$ is a geodesic with constant speed $|\sigma(s)| = 1$. Therefore:

$$D_t T = D_t \partial_t \Gamma = 0 \quad \text{and} \quad \langle T, T \rangle = |\partial_t \Gamma|^2 = 1$$

We want to show that $\langle S, T \rangle = \langle \partial_s \Gamma, \partial_t \Gamma \rangle$ is independent of t. Taking the derivative with respect to t:

$$\frac{\partial}{\partial t} \langle \partial_s \Gamma, \partial_t \Gamma \rangle = \langle D_t \partial_s \Gamma, \partial_t \Gamma \rangle + \langle \partial_s \Gamma, D_t \partial_t \Gamma \rangle$$

By the Symmetry Lemma ($D_t \partial_s \Gamma = D_s \partial_t \Gamma$) and the fact that $D_t \partial_t \Gamma = 0$:

$$\frac{\partial}{\partial t} \langle S, T \rangle = \langle D_s \partial_t \Gamma, \partial_t \Gamma \rangle + 0 = \frac{1}{2} \frac{\partial}{\partial s} \langle \partial_t \Gamma, \partial_t \Gamma \rangle = \frac{1}{2} \frac{\partial}{\partial s} |\sigma(s)|^2 = \frac{1}{2} \frac{\partial}{\partial s}(1) = 0$$

This shows $\langle \partial_s \Gamma, \partial_t \Gamma \rangle$ is constant along the radial geodesic. To evaluate this constant, we look at $t = 0$. At $t = 0$, $\Gamma(s, 0) = \exp_p(0) = p$ for all s. Thus, the derivative with respect to s at $t = 0$ is the zero vector: $\partial_s \Gamma(s, 0) = 0$. Therefore, for all t:

$$\langle \partial_s \Gamma, \partial_t \Gamma \rangle|_t = \langle \partial_s \Gamma, \partial_t \Gamma \rangle|_{t=0} = \langle 0, \partial_t \Gamma \rangle = 0$$

Conclusion: $\langle \partial_r, \partial_\theta \rangle = 0$. □

Remark 11.7. In geodesic polar coordinates, the metric tensor takes the form:

$$g = dr^2 + \sum_{i,j=1}^{n-1} h_{ij}(r, \theta) d\theta^i d\theta^j$$

There are no $dr d\theta^i$ cross-terms because $\langle \partial_r, \partial_{\theta^i} \rangle = 0$, and the coefficient of dr^2 is 1 because radial geodesics have unit speed.

11.4 Geodesics are Locally Length-Minimizing

Proposition 11.8. *If $p \in M$ and q is a point in a geodesic ball around p, then the radial geodesic from p to q is the unique shortest curve from p to q.*

Proof. Let B be a normal neighborhood (geodesic ball) centered at p. The radial geodesic $\gamma_{p,q}$ connects p to q. Let R be the length of this radial geodesic (so $q = \exp_p(R\theta_0)$). Let $\sigma : [a, b] \to M$ be any other parameterized curve connecting p to q. Assume for now that σ lies entirely within B. In geodesic polar coordinates, we can write $\sigma(t)$ as:

$$\sigma(t) = \exp_p(r(t)\theta(t))$$

The velocity vector of σ is:

$$\dot{\sigma}(t) = \dot{r}(t)\partial_r + X(t)$$

where $X(t) = \sum \dot{\theta}^i(t)\partial_{\theta^i}$ is a vector tangent to the geodesic sphere $S_r(p)$. By the Gauss Lemma, $\partial_r \perp X(t)$. Thus, using the Pythagorean theorem for the inner product:

$$|\dot{\sigma}(t)|^2 = |\dot{r}(t)\partial_r|^2 + |X(t)|^2 = \dot{r}(t)^2 + |X(t)|^2 \geq \dot{r}(t)^2$$

Taking the square root, we have $|\dot{\sigma}(t)| \geq \dot{r}(t)$. The length of σ is:

$$L(\sigma) = \int_a^b |\dot{\sigma}(t)|\, dt \geq \int_a^b \dot{r}(t)\, dt = r(b) - r(a) = R - 0 = R$$

Hence, $L(\sigma) \geq R = L(\gamma_{p,q})$, proving the radial geodesic is minimizing.

Uniqueness: For equality to hold, $L(\sigma) = R$, we must have $|\dot{\sigma}(t)| = \dot{r}(t)$ almost everywhere. This implies two conditions:

1. $|X(t)| \equiv 0$, meaning $\dot{\theta}(t) = 0$, so $\theta(t)$ is constant. The curve must be a radial ray.
2. $\dot{r}(t) \geq 0$, meaning the curve never backtracks toward p.

Thus, the only curve achieving length R is the radial geodesic itself (up to monotonic reparameterization). □

Corollary 11.9. *1. A geodesic ball around p coincides with a metric ball around p (using the Riemannian distance function).*

2. Geodesics are locally length-minimizing: every point has a neighborhood such that any two points in the neighborhood are joined by a unique length-minimizing geodesic.

12 Lecture 12

12.1 Geodesics as Locally Minimizing Curves

Proposition 12.1. *A geodesic $\gamma : I \to M$ is locally minimizing. i.e., there exists a neighborhood U of t_0 such that for any $t_1, t_2 \in U$, the restriction $\gamma|_{[t_1,t_2]}$ is the unique shortest curve connecting $\gamma(t_1)$ and $\gamma(t_2)$.*

Proof. Take W, a uniformly normal neighborhood of $\gamma(t_0)$ with respect to parameter δ. By definition, for every $p \in W$, there exists a normal δ-ball around p, and W is completely contained within that normal ball.

Let U be the connected component of $\gamma^{-1}(W)$ containing t_0. For any $t_1, t_2 \in U$, we have $\gamma(t_1), \gamma(t_2) \in W$.

Because W is uniformly normal, we can center a normal δ-ball at $\gamma(t_1)$, and we are guaranteed that $\gamma(t_2)$ lies inside this normal ball. Within a normal geodesic ball, any point can be connected to the center by a unique radial geodesic, which is strictly shorter than any other curve connecting the two points. Since the original curve $\gamma|_{[t_1,t_2]}$ is a geodesic connecting $\gamma(t_1)$ and $\gamma(t_2)$ and lies within this normal ball, it must coincide with this minimizing radial geodesic (up to parameterization). Hence, $\gamma|_{[t_1,t_2]}$ minimizes the distance. □

12.2 Geodesic Completeness and the Hopf-Rinow Theorem

Definition 12.2. A Riemannian manifold M is **geodesically complete** if every maximal geodesic is defined for all $t \in \mathbb{R}$.

Example 12.3 (Counterexample)**.** $\mathbb{R}^2 \backslash \{(0,0)\}$ is not geodesically complete. A geodesic heading directly towards the removed origin cannot be extended for all $t \in \mathbb{R}$.

Theorem 12.4 (Hopf-Rinow Theorem - Part I)**.** *Let M be a connected Riemannian manifold. M is metrically complete (as a metric space) if and only if M is geodesically complete.*

Proof. $\Leftarrow$: Suppose (M, g) is metric-complete but not geodesic-complete. Then there exists a unit-speed maximal geodesic $\gamma : [a, b) \to M$ with $b < \infty$.

Since γ is unit-speed, the distance between any two points on the curve is bounded by the arc length:

$$d(\gamma(t), \gamma(t')) \leq |t - t'|$$

Take a sequence $t_j \to b$. Since $\{t_j\}$ is a Cauchy sequence in $\mathbb{R}$, the inequality above implies that $\{\gamma(t_j)\}$ is a Cauchy sequence in M.

Because M is assumed to be metrically complete, this Cauchy sequence must converge to some point $q \in M$, i.e., $\gamma(t_j) \to q$.

Now, take a uniformly normal neighborhood W of q with respect to some radius $\delta > 0$. For sufficiently large j, we have $\gamma(t_j) \in W$. By the property of the exponential map at $\gamma(t_j)$, we can extend the geodesic γ forward from $\gamma(t_j)$ by at least a distance of δ. However, since $t_j \to b$, we can choose j such that $b - t_j < \delta/2$. The extension will then go up to $t_j + \delta > b$, which contradicts the assumption that $[a, b)$ is the maximal domain of γ.

Hence, no such finite b can exist, and M is geodesically complete.

To proof the contrary, we will use the cliam below.

Claim: If $\exp_p$ is defined on all of T_pM for some $p \in M$, then for all $q \in M$, there exists a minimal geodesic connecting p to q.

Suppose $\{q_i\}$ is a Cauchy sequence in M. Assuming the claim holds, we can write $q_i = \exp_p(V_i)$ for some $V_i \in T_pM$, where the geodesic $\gamma_i(t) = \exp_p(tV_i)$ minimizes the distance, so $|V_i|_g = d(p, q_i)$.

Since $\{q_i\}$ is a Cauchy sequence, it is bounded in M. Therefore, the distances $d(p, q_i)$ are bounded, meaning there exists $D > 0$ such that $|V_i| \leq D$ for all i.

Because the sequence $\{V_i\}$ lies in a closed bounded set in the finite-dimensional vector space T_pM, by the Bolzano-Weierstrass theorem, there exists a convergent subsequence $V_{i(k)} \to W \in T_pM$.

Since the exponential map $\exp_p : T_pM \to M$ is continuous, we have:

$$\exp_p(V_{i(k)}) \to \exp_p(W) \quad \text{in the manifold topology.}$$

Because the original sequence $\{q_i\}$ is Cauchy and a subsequence converges to $\exp_p(W)$, the entire sequence must converge to the same limit. Hence, $\lim_{i\to\infty} q_i = \exp_p(W)$ in the metric topology. This proves M is a complete metric space. $\square$

Pf of the claim.
Obs: If $\gamma : [0,T] \to M$ is a minimizing geodesic from p to q. Then for all $b \in [0,T]$, $d(p,\gamma(b)) + d(\gamma(b),q) = d(p,q)$.
We call $\gamma : [0,b] \to M$ a <u>geodesic segment</u> with $\gamma(0) = p$ and we say γ <u>aims towards q</u> if γ is maximal between p and $\gamma(b)$, and $d(p,\gamma(b)) + d(\gamma(b),q) = d(p,q)$.
So its enough to show there is such a geodesic $\gamma : p \to \gamma(b)$ that towards for q with

$$L(\gamma) = d(p,q). \quad (= L(\gamma) + d(\gamma(b),q) \implies \gamma(b) = q)$$

To show this, take $\varepsilon > 0$. so $\overline{B_\varepsilon(p)}$ is a closed normal ball.
If $q \in \overline{B_\varepsilon(p)}$. then $\exists$ minimizing geodesic from p to q. Done.
If $q \notin \overline{B_\varepsilon(p)}$. Choose $x \in \partial B_\varepsilon(p)$ which minimizes the distance to q. We have unit-speed geodesic $\gamma : p \to x$. and we extend it to all $t \in \mathbb{R}$.
Since $d(p,q) = d(p,x) + d(x,q)$, $\gamma|_{[0,\varepsilon]}$ aims towards q.
Put $T = d(p,q)$, $\mathcal{A} = \{b \in [0,T] : \gamma|_{[0,b]} \text{ aims towards } q\}$.
$\mathcal{A}$ is a closed subset of $[0,T]$. Let $A = \max \mathcal{A}$.
If $A = T$. done. otherwise put $y = \gamma(A) \neq q$.
Take normal δ-Ball $\overline{B_\delta(y)}$ so $q \notin \overline{B_\delta(y)}$. Take a geodesic τ from y to $z \in \partial B_\delta(y)$. so τ aims towards q.

$$\begin{aligned} d(p,z) &\geq d(p,q) - d(z,q) && \text{So } \gamma|_{[0,A]} \cup \tau \text{ is minimizing curve } p \to z. \\ &= d(p,y) + d(y,z). && \text{hence its geodesic, aims towards } q. \end{aligned}$$

Corollaries of Hopf-Rinow:

1. If $\exists p \in M$ such that $\exp_p$ is defined on all of T_pM, then M is complete (metrically and geodesically).
2. If M is complete, then any two points can be connected by a minimizing geodesic.
3. M is Compact $\implies$ M is Geodesically complete. (Since compact metric spaces are complete).
4. If M is complete, $\exp_p : T_pM \to M$ is surjective for all $p \in M$.

12.3 Covering Maps and Local Isometries

Theorem 12.5. *Suppose $\tilde{M}, M$ are connected Riemannian manifolds, $\tilde{M}$ is complete, and $\pi : \tilde{M} \to M$ is a local isometry. Then M is complete, and π is a covering map.*

Recall: A map $\pi : \tilde{M} \to M$ is a covering map if it is surjective and for every $p \in M$, there exists a neighborhood U of p such that $\pi^{-1}(U) = \coprod_\alpha V_\alpha$ (disjoint union of open sets), and the restriction $\pi|_{V_\alpha} : V_\alpha \to U$ is a diffeomorphism.

Example 12.6 (Counterexample)**.** Let $\tilde{M} = (0, 10) \subset \mathbb{R}$, $M = S^1$, and $\pi(x) = e^{ix}$. Here, π is a local isometry but **not** a covering map. This is because $\tilde{M}$ is not complete.

Lemma 12.7 (Geodesic Lifting Lemma)**.** *If $p \in M$, $\tilde{p} \in \pi^{-1}(p)$, and $\gamma : I \to M$ is a geodesic, then there exists a unique geodesic $\tilde{\gamma}$ starting at $\tilde{p}$ such that $\pi \circ \tilde{\gamma} = \gamma$.*

Proof. Let $V = \gamma'(0) \in T_pM$. Since π is a local isometry, the differential $d\pi_{\tilde{p}} : T_{\tilde{p}}\tilde{M} \to T_pM$ is an isometric isomorphism. We can put $\tilde{V} = (d\pi_{\tilde{p}})^{-1}(V) \in T_{\tilde{p}}\tilde{M}$.

Define $\tilde{\gamma}(t) = \exp_{\tilde{p}}(t\tilde{V})$. Because local isometries map geodesics to geodesics, $\pi \circ \tilde{\gamma}$ is a geodesic in M with initial position $\pi(\tilde{p}) = p$ and initial velocity $d\pi_{\tilde{p}}(\tilde{V}) = V$. By the uniqueness of geodesics with given initial conditions, we must have $\pi \circ \tilde{\gamma} = \gamma$. □

Proof of the Theorem. **1. M is complete:** Choose any $p \in \pi(\tilde{M})$. Let γ be a geodesic in M starting at p. Let $\tilde{\gamma}$ be its lifted geodesic in $\tilde{M}$ starting at some $\tilde{p} \in \pi^{-1}(p)$. Since $\tilde{M}$ is complete, $\tilde{\gamma}$ is defined for all $t \in \mathbb{R}$. Consequently, $\pi \circ \tilde{\gamma} = \gamma$ is defined for all $t \in \mathbb{R}$. Thus, M is geodesically complete.

2. π is surjective: Choose any $\tilde{p} \in \tilde{M}$ and let $p = \pi(\tilde{p})$. Let $q \in M$ be any other point ($q \neq p$). Since we proved M is complete, by the Hopf-Rinow theorem, there exists a minimizing unit-speed geodesic $\gamma : p \to q$.

Let $d = d(p, q)$, so $\gamma(d) = q$. Lift γ to a geodesic $\tilde{\gamma}$ in $\tilde{M}$ starting at $\tilde{p}$. Because $\tilde{M}$ is complete, $\tilde{\gamma}$ is defined for all t, including $t = d$. Let $\tilde{q} = \tilde{\gamma}(d)$. We then have:

$$q = \gamma(d(p, q)) = \pi \circ \tilde{\gamma}(d(p, q)) = \pi(\tilde{q})$$

Thus, $q \in \mathrm{Im}(\pi)$, proving that π is surjective. (The property of being a covering map then follows from standard arguments utilizing the fact that geodesics lift uniquely and exp is a local diffeomorphism). □

13 Lecture 13

13.1 Distance function

Say M is a connected Riemannian manifold, $S \subset M$ any subset.

Definition 13.1. $d(x, S) = \inf_{p \in S} d(x, p)$.

Lemma 13.2. *$|d(x, S) - d(y, S)| \leq d(x, y)$, i.e., $d(\cdot, S)$ is Lipschitz.*

Fact: Any Lipschitz function on a Riemannian manifold is differentiable almost everywhere (a.e.). If it is differentiable, $|\nabla f(x)| \leq 1$.

Proposition 13.3. *If $f(\cdot) = d(\cdot, S)$ is C^1 on an open subset U of $M \setminus S$, then $|\nabla f(x)| = 1$ on U.*

Proof. Suppose $|\nabla f| < 1$, then $\exists \epsilon, \delta > 0$ so on $\overline{B_\epsilon(x)} \subset U$, $|\nabla f| \leq 1 - \delta$.

Choose $c < \delta\epsilon$. $\exists$ admissible curve $\alpha : [0, b] \to M$ of unit speed, with $\alpha(0) = x$, $\alpha(b) = q \in S$, and $b = L(\alpha) < d(x, S) + c$.

Since $\overline{B_\epsilon(x)} \subset U \subset M \setminus S$, we must have $b > \epsilon$. So $\alpha|_{[\epsilon, b]}$ is an admissible curve. We have:

$$d(\alpha(\epsilon), S) \leq d(\alpha(\epsilon), q) \leq L(\alpha|_{[\epsilon,b]}) < d(x, S) + c - \epsilon.$$

On the other hand,

$$\left| \frac{d}{dt} f(\alpha(t)) \right| = |\langle \nabla f, \alpha' \rangle| \leq 1 - \delta \quad \text{for } t \in [0, \epsilon].$$

Integrating this, we get:

$$f(x) - f(\alpha(\epsilon)) \leq (1 - \delta)\epsilon.$$

Recalling $f(\cdot) = d(\cdot, S)$, we put $t = \epsilon$ to obtain $d(x, S) - d(\alpha(\epsilon), S) \leq (1 - \delta)\epsilon$. So,

$$d(\alpha(\epsilon), S) \geq d(x, S) - \epsilon + \delta\epsilon > d(x, S) - \epsilon + c.$$

This contradicts the earlier inequality $d(\alpha(\epsilon), S) < d(x, S) + c - \epsilon$. □

Definition 13.4. If $U \subset M$ is open, a **local distance function** on U is a C^1 function so that $|\nabla f| = 1$.

If f is a local distance function, then f is locally the distance from a level set of f.

Theorem 13.5. *If f is a local distance function, then $\nabla_{\nabla f} \nabla f = 0$. Each integral curve of ∇f is a geodesic.*

Proof. Put $F = \nabla f$. For any vector field W, $Wf = df(W) = \langle F, W \rangle$. And $Ff = |F|^2 = 1$.

We compute:

$$\begin{aligned}
\langle W, \nabla_F F \rangle &= F\langle W, F \rangle - \langle \nabla_F W, F \rangle \\
&= F(Wf) - \langle \nabla_W F + [F, W], F \rangle \quad \text{(since } \nabla \text{ is torsion-free)} \\
&= F(Wf) - [F, W]f - \frac{1}{2} W\langle F, F \rangle \\
&= W(Ff) - \frac{1}{2} W(1) \\
&= W(1) - 0 = 0.
\end{aligned}$$

Hence $\nabla_F F = 0$. If $\gamma' = \nabla f$, then γ is a unit-speed geodesic. □

13.2 Riemann Covering

Theorem 13.6 (Ambrose)**.** *Say $\tilde{M}, M$ are connected manifolds, $\tilde{M}$ is complete, and $\pi : \tilde{M} \to M$ is a local isometry. Then M is complete and π is a covering map.*

Proof. Use the lifting lemma. Let γ be a geodesic through p. For $\tilde{p} \in \pi^{-1}(p)$, then $\exists \tilde{\gamma}$ geodesic through $\tilde{p}$ such that $\pi \circ \tilde{\gamma} = \gamma$. This shows M is complete and π is surjective.

For the second assertion, let $p \in M$ and let $U = B_\epsilon(p)$ be a normal ϵ-ball. Write $\pi^{-1}(p) = \{\tilde{p}_\alpha\}$. Say U_α is the ϵ-ball around $\tilde{p}_\alpha$. I claim that for $\alpha \neq \beta$, $U_\alpha \cap U_\beta = \emptyset$.

- *Proof of claim:* Draw minimal geodesic $\tilde{\gamma} : \tilde{p}_\alpha \to \tilde{p}_\beta$. Then $\pi \circ \tilde{\gamma}$ is a non-constant geodesic from $p \to p$. So $L(\tilde{\gamma}) = L(\pi \circ \tilde{\gamma}) \geq 2\epsilon$ (since it has to go outside the ϵ-ball!). This proves the claim.

To show $\pi^{-1}(U) = \coprod_\alpha U_\alpha$:

a. Show $\pi^{-1}(U) \supset U_\alpha$: Say $\tilde{q}_\alpha \in U_\alpha$. Let $\tilde{\gamma}_\alpha$ be the geodesic $\tilde{p}_\alpha \to \tilde{q}_\alpha$ of length $< \epsilon$. Then $\pi \circ \tilde{\gamma}_\alpha$ is a geodesic in U, starting at p and of length $< \epsilon$. Hence $\pi(\tilde{q}_\alpha) \in U$, i.e., $\tilde{q}_\alpha \in \pi^{-1}(U)$.

b. Show $\pi^{-1}(U) \subset \coprod_\alpha U_\alpha$: Say $\tilde{q} \in \pi^{-1}(U)$. Put $q = \pi(\tilde{q}) \in U$. Let γ be the geodesic $p \to q$ of length $< \epsilon$. Lift this to $\tilde{\gamma}$ starting from $\tilde{q}$; the endpoint of $\tilde{\gamma}$ is some $\tilde{p}_\alpha \in \pi^{-1}(p)$. Hence $\tilde{q} \in U_\alpha$.

To show $\pi|_{U_\alpha} : U_\alpha \to U$ is a diffeomorphism: $\pi|_{U_\alpha}$ is a local diffeomorphism. $(\pi|_{U_\alpha})^{-1}$ is a local diffeomorphism by lifting radial geodesics. □

Corollary 13.7. *If $M, \tilde{M}$ are connected, and $\pi : \tilde{M} \to M$ is a Riemannian covering, then M is complete iff $\tilde{M}$ is complete.*

Proof. π is a local isometry, so $\tilde{M}$ complete $\implies$ M complete. Conversely, suppose M is complete. For $\tilde{p} \in \tilde{M}$, $\tilde{V} \in T_{\tilde{p}}\tilde{M}$, put $p = \pi(\tilde{p})$, $V = d\pi_{\tilde{p}}(\tilde{V})$. $\exists$ geodesic γ with $\gamma(0) = p, \gamma'(0) = V$. Lift to $\tilde{\gamma}$. $|\tilde{\gamma}'(0)| = |\tilde{V}|$. Hence all geodesics in $\tilde{M}$ are defined for $t \in \mathbb{R}$. □

Corollary 13.8. *Suppose (M, g) is a complete Riemannian manifold, $p, q \in M$, and $\mathcal{C}$ is a homotopy class of curves $p \to q$. Then $\exists$ a minimal geodesic in $\mathcal{C}$.*

Proof. Put $\tilde{M}$ to be the universal cover. Fix $\tilde{p} \in \pi^{-1}(p)$. Then $\mathcal{C} \cong \pi^{-1}(q)$. Given $\tilde{q}_\alpha \in \pi^{-1}(q)$, $\exists$ minimal geodesic $\tilde{\gamma} : \tilde{p} \to \tilde{q}_\alpha$. Put $\gamma = \pi \circ \tilde{\gamma}$. □

Corollary 13.9. *For any $\alpha \in \pi_1(M, p)$, $\exists$ minimal length loop geodesic in α.*

Remark 13.10. The geodesic loop could have an angle at p.

We have a correspondence:

$$\{\text{free homotopy classes in } M\} \longleftrightarrow \{\text{conjugacy classes of } \pi_1(M)\}.$$

We can minimize length in a free homotopy class and get a closed smooth geodesic.

Corollary 13.11. *If $\pi_1(M) \neq \{e\}$, then M has a nonconstant smooth closed geodesic.*

Fact:[Lusternik-Schnirelmann] For $n \geq 2$, if $M \cong S^n$, there exist at least three distinct "prime" closed geodesics.

14 Lecture 14

14.1 Riemann Curvature Tensor

Definition 14.1. The Riemann curvature tensor is defined as

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \in \mathfrak{X}(M).$$

Proposition 14.2. *It is a* $(1,3)$ *tensor field.*

Remark 14.3. *Notation:* $R(\partial_i, \partial_j)\partial_k = \sum_l R^l_{ijk}\partial_l$. As a tensor, $R = \sum_{i,j,k,l} R^l_{ijk} dx^i \otimes dx^j \otimes dx^k \otimes \partial_l$.

Definition 14.4. $Rm(X,Y,Z,W) = \langle R(X,Y)Z, W\rangle$. Rm is a $(0,4)$ tensor field, with $R_{ijkl} = \sum_m R^m_{ijk} g_{ml}$.

Remark 14.5. *Coordinate formula:*

$$R^l_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \sum_m \left(\Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik}\right).$$

Remark 14.6. *Change of coordinates:*

$$\tilde{g}_{ij}(y) = \sum_{k,l} g_{kl}(x) \frac{\partial x^k}{\partial y^i}\frac{\partial x^l}{\partial y^j},$$

$$\tilde{R}_{ijkl}(y) = \sum_{a,b,c,d} R_{abcd}(x) \frac{\partial x^a}{\partial y^i}\frac{\partial x^b}{\partial y^j}\frac{\partial x^c}{\partial y^k}\frac{\partial x^d}{\partial y^l}.$$

Remark 14.7. *Riemann's reasoning for* $R(X,Y)Z$*:*

1. Curvature at x should depend on $g_{ij}, g^{ij}, \partial_k g_{ij}, \partial_k \partial_l g_{ij}$.
2. If $g_{ij} = \frac{\partial \vec{X}}{\partial x^i} \cdot \frac{\partial \vec{X}}{\partial x^j}$ for some $\vec{X} : \mathbb{R}^n \to \mathbb{R}^N$, then the curvature should be zero.

Proposition 14.8 (Symmetries)**.**

1. $Rm(W,X,Y,Z) = -Rm(X,W,Y,Z) \implies R_{ijkl} = -R_{jikl}$.
2. $Rm(W,X,Y,Z) = -Rm(W,X,Z,Y) \implies R_{ijkl} = -R_{ijlk}$.
3. $Rm(W,X,Y,Z) = Rm(Y,Z,W,X) \implies R_{ijkl} = R_{klij}$.
4. ***1st Bianchi identity:***

$$Rm(W,X,Y,Z) + Rm(X,Y,W,Z) + Rm(Y,W,X,Z) = 0 \implies R_{ijkl} + R_{jkil} + R_{kijl} = 0.$$

Proposition 14.9 (2nd Bianchi Identity)**.** *Define* $(\nabla_X Rm)(Y,Z,V,W) = \nabla Rm(X,Y,Z,V,W)$*. Then:*

$$\nabla Rm(X,Y,Z,V,W) + \nabla Rm(X,Y,V,W,Z) + \nabla Rm(X,Y,W,Z,V) = 0.$$

In coordinates (cyclic permutation on the last three indices):

$$R_{ijkl;m} + R_{ijlm;k} + R_{ijmk;l} = 0.$$

Example 14.10.

- $n = 1$: $R_{1111} = 0$, so $Rm = 0$.
- $n = 2$: Say $\{e_1, e_2\}$ is an ONB for T_pM, $R_{1221} = K$ (Gaussian curvature on M).
- n: There are $\frac{n^2(n^2-1)}{12}$ independent components of Rm.

Definition 14.11 (Ricci Curvature)**.** The Ricci curvature is a $(0,2)$ tensor field defined by:

$$R_{ij} = \sum_{k,l} g^{kl} R_{iklj}, \quad R_{ij} = R_{ji}.$$

Definition 14.12 (Scalar Curvature)**.** The scalar curvature is defined by:

$$S = \sum_{i,j} g^{ij} R_{ij} = \sum_{i,j,k,l} g^{ij} g^{kl} R_{ikjl}.$$

14.2 Sectional Curvature & Schur's Theorem

Proposition 14.13 (Contracted Bianchi Identity)**.**

$$R_{ij}{}^{;j} = \frac{1}{2} S_{;i} \quad \textit{i.e.,} \quad \left(R_{ij} - \frac{1}{2} S g_{ij} \right)^{;j} = 0.$$

Here the notation means $S_{;i} = \nabla_i S$, $R_{ij;k} = \nabla_k R_{ij}$.

Definition 14.14 (Pointwise Constant Sectional Curvature)**.** M has pointwise constant sectional curvature if $\exists K \in C^\infty(M)$ such that

$$R_{ijkl} = K(g_{ik} g_{jl} - g_{il} g_{jk}).$$

Theorem 14.15 (Schur's Theorem)**.** *If M is connected with* $\dim M > 2$ *and has pointwise constant sectional curvature, then K is constant.*

Proof. Given $R_{ijkl} = K(g_{ik} g_{jl} - g_{il} g_{jk})$. By index substitution, $R_{ikjl} = K(g_{ij} g_{kl} - g_{il} g_{kj})$. Taking the trace with g^{kl} yields the Ricci tensor:

$$\begin{aligned} R_{ij} &= g^{kl} R_{ikjl} = K g^{kl} (g_{ij} g_{kl} - g_{il} g_{kj}) \\ &= K(g_{ij} \cdot n - g_{ij}) = K(n-1) g_{ij}. \end{aligned}$$

Taking the trace again gives the scalar curvature:

$$S = g^{ij} R_{ij} = K(n-1) \cdot n = Kn(n-1).$$

Now apply the contracted Bianchi identity, $R_{ij}{}^{;j} = \frac{1}{2} S_{;i}$:

$$\begin{aligned} &\text{LHS:} \quad R_{ij}{}^{;j} = \nabla^j (K(n-1) g_{ij}) = (n-1) K_{;j} \delta_i^j = (n-1) K_{;i}. \\ &\text{RHS:} \quad \frac{1}{2} S_{;i} = \frac{1}{2} n(n-1) K_{;i}. \end{aligned}$$

Equating both sides:

$$\frac{1}{2} n(n-1) K_{;i} = (n-1) K_{;i} \implies (n-1) \left(\frac{n}{2} - 1 \right) K_{;i} = 0.$$

Since $n > 2$, we have $(n-1) \neq 0$ and $\left(\frac{n}{2} - 1\right) \neq 0$. Thus, $K_{;i} = 0$, meaning K is locally constant. Since M is connected, K is globally constant. □

Definition 14.16 (Einstein Manifold)**.** An Einstein manifold is a Riemannian manifold such that $R_{ij} = c g_{ij}$ for some constant $c \in \mathbb{R}$.

Proposition 14.17. *If M is connected with* $\dim M > 2$ *and $R_{ij} = f g_{ij}$ for some $f \in C^\infty(M)$, then f is constant. (The proof follows identically from the contracted Bianchi identity as in Schur's Theorem.)*

14.3 Isometric Immersions

Definition 14.18 (Isometric Immersion)**.** Let (M, g) and $(\tilde{M}, \tilde{g})$ be Riemannian manifolds. An immersion $i : M \to \tilde{M}$ is an isometric immersion if $g = i^*\tilde{g}$, i.e.,

$$g(X, Y) = \tilde{g}(di(X), di(Y)) \quad \text{for all } X, Y \in T_pM.$$

Given $\tilde{g}$ and an immersion i, there is a unique metric g on M such that i is an isometric immersion.

Definition 14.19 (Normal Bundle)**.** The tangent space of the ambient manifold at p splits as $T_{i(p)}\tilde{M} = di_p(T_pM) \oplus T_pM^\perp$. The normal bundle is $NM = \coprod_{p\in M} T_pM^\perp$, so $T\tilde{M}|_M \simeq TM \oplus NM$.

Remark 14.20. With $\tilde{g}$, we can define the orthogonal projections:

$$\Pi^\top : T\tilde{M}|_M \to TM,$$
$$\Pi^\perp : T\tilde{M}|_M \to NM.$$

14.4 Induced Connection on Submanifolds

Say M is embedded in $\tilde{M}$.

Proposition 14.21. *Given $X, Y \in \mathfrak{X}(M)$, extend to $\tilde{X}, \tilde{Y}$ on a neighborhood of M in $\tilde{M}$. Then*

$$\pi^\top(\tilde{\nabla}_{\tilde{X}}\tilde{Y}) = \nabla_X Y,$$

the Levi-Civita connection on M.

Proof. Put $D_XY = \pi^\top(\tilde{\nabla}_{\tilde{X}}\tilde{Y})$. Show D_XY is independent of extensions. D is a connection, metric-compatible and torsion-free. Then $D_XY = \nabla_XY$. □

2nd Fundamental Form

Definition 14.22.

$$\mathrm{II}(X, Y) = \pi^\perp(\tilde{\nabla}_{\tilde{X}}\tilde{Y}),$$

a section of NM (the normal bundle).

Proposition 14.23.

1. *This is independent of extensions $\tilde{X}, \tilde{Y}$.*
2. *It's $C^\infty(M)$-linear in X, Y.*
3. $\mathrm{II}(X, Y) = \mathrm{II}(Y, X)$.

Proofs of the Properties

Proof of 3.

$$\begin{aligned}\mathrm{II}(X, Y) - \mathrm{II}(Y, X) &= \pi^\perp(\tilde{\nabla}_{\tilde{X}}\tilde{Y} - \tilde{\nabla}_{\tilde{Y}}\tilde{X}) \\ &= \pi^\perp([\tilde{X}, \tilde{Y}]) \\ &= 0 \quad (\text{since } [\tilde{X}, \tilde{Y}]|_M = [X, Y] \text{ is tangent to } M)\end{aligned}$$

□

Proof of 1. Using 3, enough to show independent of choice of $\tilde{X}$.

$$\pi^\perp(\tilde{\nabla}_{\tilde{X}}\tilde{Y})(p) \quad \text{depends on } \tilde{X} \text{ through } \tilde{X}(p) = X(p).$$

□

Proof of 2. Using 3, enough to show it's $C^\infty(M)$-linear in X. Let $\tilde{f}$ be an extension of $f \in C^\infty(M)$.

$$\begin{aligned}
\mathrm{I\!I}(fX, Y) &= \pi^\perp(\tilde{\nabla}_{\tilde{f}\tilde{X}}\tilde{Y}) \\
&= \pi^\perp(\tilde{f}\tilde{\nabla}_{\tilde{X}}\tilde{Y}) \\
&= f\pi^\perp(\tilde{\nabla}_{\tilde{X}}\tilde{Y}) \\
&= f\mathrm{I\!I}(X, Y)
\end{aligned}$$

□

Summary (Gauss Formula)

Given an immersion, we get:

$$\mathrm{I\!I}_p : T_pM \times T_pM \to N_pM$$

which is symmetric in its arguments.

On M, we have the decomposition (Gauss Formula):

$$\tilde{\nabla}_{\tilde{X}}\tilde{Y} = \nabla_X Y + \mathrm{I\!I}(X, Y)$$

15 Lecture 15

15.1 Proof of Symmetries of Curvature Tensors

Let $\{E_i\}$ be a local frame, and $\{\varepsilon^i\}$ be the dual coframe.

$$\begin{aligned} d\varepsilon^i &= \varepsilon^j \wedge \omega^i_j + \tau^i \quad \text{(Definition of torsion } \tau\text{)} \\ g_{jk}\omega^k_i + g_{ik}\omega^k_j &= dg_{ij} \quad \text{(Metric compatibility)} \\ \Omega^j_i &= d\omega^j_i - \omega^k_i \wedge \omega^j_k \quad \text{(Definition of curvature } \Omega\text{)} \end{aligned}$$

Suppose $\{E_i\}$ is orthonormal. Torsion vanishes, and $g_{ij} = \delta_{ij}$. Then $d\varepsilon^j = \varepsilon^i \wedge \omega^j_i$, and $\omega^i_j + \omega^j_i = 0$.

Proposition 15.1. $\Omega^j_i = -\Omega^i_j$

Proof.

$$\begin{aligned} \Omega^j_i &= d\omega^j_i - \omega^k_i \wedge \omega^j_k \\ &= -d\omega^i_j + \omega^j_k \wedge \omega^k_i \\ &= -d\omega^i_j + \omega^k_j \wedge \omega^i_k \\ &= -\Omega^i_j \end{aligned}$$

□

Corollary 1. $R_{klij} = -R_{klji}$

Pf of 1st Bianchi:

Proof. Start with $d\varepsilon^j = \varepsilon^i \wedge \omega^j_i$. Apply the exterior derivative d:

$$\begin{aligned} 0 &= d\varepsilon^i \wedge \omega^j_i - \varepsilon^i \wedge d\omega^j_i \\ &= (\varepsilon^k \wedge \omega^i_k) \wedge \omega^j_i - \varepsilon^k \wedge d\omega^j_k \\ &= \varepsilon^k \wedge \omega^i_k \wedge \omega^j_i - \varepsilon^k \wedge d\omega^j_k \\ &= -\varepsilon^k \wedge (d\omega^j_k - \omega^i_k \wedge \omega^j_i) \\ &= -\varepsilon^k \wedge \Omega^j_k \\ &= -\varepsilon^k \wedge \left(\frac{1}{2} R_{abk}{}^j \varepsilon^a \wedge \varepsilon^b\right) \end{aligned}$$

$\Rightarrow 0 = -\frac{1}{2} R_{abkj}\varepsilon^k \wedge \varepsilon^a \wedge \varepsilon^b$ (1st Bianchi). □

Pf of 2nd Bianchi:

Proof. From $\Omega^j_i = d\omega^j_i - \omega^k_i \wedge \omega^j_k$, we take the exterior derivative:

$$\begin{aligned} d\Omega^j_i &= -d\omega^k_i \wedge \omega^j_k + \omega^k_i \wedge d\omega^j_k \\ &= -(\Omega^k_i + \omega^l_i \wedge \omega^k_l) \wedge \omega^j_k + \omega^k_i \wedge (\Omega^j_k + \omega^l_k \wedge \omega^j_l) \\ &= -\Omega^k_i \wedge \omega^j_k + \omega^k_i \wedge \Omega^j_k \end{aligned}$$

$\Rightarrow d\Omega^j_i + \Omega^k_i \wedge \omega^j_k - \omega^k_i \wedge \Omega^j_k = 0$ (2nd Bianchi). □

Now, evaluate $d\Omega_i^j$:

$$\begin{aligned} d\Omega_i^j &= \varepsilon^m \wedge \nabla_{E_m} \Omega_i^j + \dots \\ &= \frac{1}{2}\varepsilon^m \wedge \nabla_{E_m}(R_{klij})\varepsilon^k \wedge \varepsilon^l + \dots \\ &= \frac{1}{2}\nabla_{E_m} R_{klij}\varepsilon^m \wedge \varepsilon^k \wedge \varepsilon^l + \dots \end{aligned}$$

Can choose an ortho frame $\{E_i\}$ near p so that $\omega_i^j(p) = 0$. To do this, choose an orthobasis $\{E_i(p)\}$ for T_pM, and parallel transport along radial geodesics from p. At p, $E_m = \nabla_m$.

Get $0 = \nabla_m R_{klij}\varepsilon^m \wedge \varepsilon^k \wedge \varepsilon^l$. This equation is independent of the choice of the orthonormal frame.

15.2 Cartan's Moving Frame

To compute curvature:

1. Choose orthonormal frame $\{E_i\}$.
2. Using $\omega_i^j + \omega_j^i = 0$ and $d\varepsilon^i = \varepsilon^j \wedge \omega_j^i$ to determine ω_i^j.
3. Compute Ω_i^j.

Example 15.2. Upper half plane. $g = \frac{dx^2+dy^2}{y^2}$. Let $\varepsilon^1 = \frac{dx}{y}$, $\varepsilon^2 = \frac{dy}{y}$.

$$\begin{aligned} &d\varepsilon^1 = -\frac{1}{y^2}dy \wedge dx, \quad d\varepsilon^2 = 0. \\ &d\varepsilon^1 = \varepsilon^2 \wedge \omega_2^1. \quad \text{Since } \varepsilon^2 \wedge \varepsilon^1 = \frac{dy}{y} \wedge \frac{dx}{y} = -\frac{1}{y^2}dy \wedge dx = d\varepsilon^1 \\ &\quad \Rightarrow \omega_2^1 = \varepsilon^1 + (?)\varepsilon^2. \end{aligned}$$

However $d\varepsilon^2 = \varepsilon^1 \wedge \omega_1^2 = \varepsilon^1 \wedge (\varepsilon^1 + (?)\varepsilon^2) = 0 \Rightarrow (?) = 0$. So $\omega_1^2 = \varepsilon^1$.

$$\begin{aligned} \Omega_1^2 &= d\omega_1^2 - \omega_1^1 \wedge \omega_1^2 - \omega_1^2 \wedge \omega_2^2 \\ &= d\omega_1^2 = d\varepsilon^1 = -\frac{1}{y^2}dy \wedge dx = \varepsilon^1 \wedge \varepsilon^2 \\ &= \frac{1}{2}R_{kl12}\varepsilon^k \wedge \varepsilon^l = R_{1212}\varepsilon^1 \wedge \varepsilon^2 \end{aligned}$$

$\Rightarrow R_{1212} = 1$ and $K = R_{2112} = -1$.

15.3 Submanifolds and the Second Fundamental Form

Let $M \subset \tilde{M}$ be a submanifold. On $\tilde{M}$, we have $\tilde{\nabla}_{\tilde{X}}\tilde{Y} = \nabla_X Y + \mathrm{II}(X, Y)$, where:

- X, Y: vector fields on M.
- $\tilde{X}, \tilde{Y}$: extensions on a neighborhood of M.
- $\nabla_X Y = \pi^\top \tilde{\nabla}_{\tilde{X}}\tilde{Y}$.
- $\mathrm{II}(X, Y) \in N_pM$ (Normal bundle).
- $\mathrm{II}(X, Y) = \pi^\perp \tilde{\nabla}_{\tilde{X}}\tilde{Y}$.
- $\mathrm{II}_p : T_pM \times T_pM \to N_pM$, with $\mathrm{II}(X, Y) = \mathrm{II}(Y, X)$.

We call $\nabla_X Y$ the intrinsic part and $\mathrm{II}(X,Y)$ the extrinsic part.

Say $\gamma : I \to M$ is a curve, and V is a vector field along γ tangent to M.

Proposition 15.3. $\tilde{D}_t V = D_t V + \mathrm{II}(\dot{\gamma}, V)$

Proof. Around $t_0 \in I$, choose a basis $\{E_i\}$ of tangential vectors along γ that are extendable. Write $V(t) = \sum V^i(t) E_i(t)$.

$$\begin{aligned}
\tilde{D}_t V &= \sum \left(\frac{dV^i}{dt} E_i + V^i \tilde{\nabla}_{\dot{\gamma}} E_i \right) \\
&= \sum \left(\frac{dV^i}{dt} E_i + V^i (\nabla_{\dot{\gamma}} E_i + \mathrm{II}(\dot{\gamma}, E_i)) \right) \\
&= D_t V + \mathrm{II}\left(\dot{\gamma}, \sum V^i E_i\right) \\
&= D_t V + \mathrm{II}(\dot{\gamma}, V)
\end{aligned}$$

□

Corollary 2. $\tilde{D}_t \dot{\gamma} = D_t \dot{\gamma} + \mathrm{II}(\dot{\gamma}, \dot{\gamma})$.

Corollary 3.

1. γ is a geodesic in $M \Leftrightarrow$ The acceleration $\tilde{D}_t \dot{\gamma}$ is normal to M.
2. γ a geodesic in $M \Rightarrow \tilde{D}_t \dot{\gamma} = \mathrm{II}(\dot{\gamma}, \dot{\gamma})$.
3. If γ is a curve in M and it's also a geodesic in $\tilde{M}$, then it's a geodesic in M and $\mathrm{II}(\dot{\gamma}, \dot{\gamma}) = 0$.

Weingarten Equation

Say X, Y are tangent to M, and $\tilde{X}, \tilde{Y}$ are extensions to $\tilde{M}$. Suppose N is a normal vector field, i.e., a section of NM. Extend N to $\tilde{N}$ as a vector field. Then $\langle \tilde{\nabla}_{\tilde{X}} N, Y \rangle = -\langle \mathrm{II}(X,Y), N \rangle$.

Proof. On M, $\langle N, Y \rangle = 0$, so $X\langle N, Y \rangle = 0$. i.e., when restricted to M:

$$\begin{aligned}
0 = \tilde{X}\langle \tilde{N}, \tilde{Y} \rangle &= \langle \tilde{\nabla}_{\tilde{X}} \tilde{N}, \tilde{Y} \rangle + \langle \tilde{N}, \tilde{\nabla}_{\tilde{X}} \tilde{Y} \rangle \\
&= \langle \tilde{\nabla}_{\tilde{X}} \tilde{N}, Y \rangle + \langle N, \tilde{\nabla}_{\tilde{X}} \tilde{Y} \rangle
\end{aligned}$$

So, $\langle \tilde{\nabla}_{\tilde{X}} \tilde{N}, Y \rangle = -\langle N, \tilde{\nabla}_{\tilde{X}} \tilde{Y} \rangle = -\langle N, \pi^{\perp} \tilde{\nabla}_{\tilde{X}} \tilde{Y} \rangle = -\langle N, \mathrm{II}(X,Y) \rangle$. □

Gauss Equation

For tangential vector fields X, Y, Z, W:

$$\tilde{Rm}(X,Y,Z,W) = Rm(X,Y,Z,W) - \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z) \rangle + \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W) \rangle$$

Proof.

$$\begin{aligned}
\tilde{Rm}(X,Y,Z,W) &= \langle \tilde{\nabla}_{\tilde{X}} \tilde{\nabla}_{\tilde{Y}} \tilde{Z} - \tilde{\nabla}_{\tilde{Y}} \tilde{\nabla}_{\tilde{X}} \tilde{Z} - \tilde{\nabla}_{[\tilde{X},\tilde{Y}]} \tilde{Z}, W \rangle \\
&= \langle \tilde{\nabla}_{\tilde{X}} (\nabla_Y Z + \mathrm{II}(Y,Z)) - \cdots - \nabla_{[X,Y]} Z - \mathrm{II}([X,Y], Z), W \rangle \\
&= \langle \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z, W \rangle \\
&\quad + \langle \tilde{\nabla}_{\tilde{X}} \mathrm{II}(Y,Z), W \rangle - \langle \tilde{\nabla}_{\tilde{Y}} \mathrm{II}(X,Z), W \rangle - \langle \mathrm{II}([X,Y], Z), W \rangle \\
&= Rm(X,Y,Z,W) + \langle \tilde{\nabla}_{\tilde{X}} \mathrm{II}(Y,Z), W \rangle - \langle \tilde{\nabla}_{\tilde{Y}} \mathrm{II}(X,Z), W \rangle - 0
\end{aligned}$$

By the Weingarten equation, $\langle \tilde{\nabla}_{\tilde{X}} \mathrm{II}(Y,Z), W \rangle = -\langle \mathrm{II}(Y,Z), \mathrm{II}(X,W) \rangle$. Substituting this yields:

$$\tilde{Rm}(X,Y,Z,W) = Rm(X,Y,Z,W) - \langle \mathrm{II}(X,W), \mathrm{II}(Y,Z) \rangle + \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W) \rangle$$

□

15.4 Hypersurfaces in Euclidean Space

From now on, look at $M^n \subset \mathbb{R}^{n+1}$ where $\dim M = n$.

- M is orientable $\Leftrightarrow$ unit section N of NM is defined up to ± 1.
- If M is not orientable, $\exists$ unit normal field N locally.

Definition 15.4. $h(X,Y) = \langle \mathrm{II}(X,Y), N\rangle$. The 2nd fundamental form. i.e., $\mathrm{II}(X,Y) = h(X,Y)N$.

The shape operator S is the section of $\mathrm{End}(TM)$ s.t. $\langle SX, Y\rangle = h(X,Y)$.

Lemma 15.5. *S on T_pM is self-adjoint. $S^i_j = g^{ik}h_{kj}$.*

We have $\tilde{\nabla}_{\tilde{X}}\tilde{Y} = \nabla_X Y + h(X,Y)N$.
For the Weingarten equation: $\tilde{\nabla}_{\tilde{X}}N = -SX$.

Proof. $\langle \tilde{\nabla}_{\tilde{X}}N, Y\rangle = -\langle N, \tilde{\nabla}_{\tilde{X}}Y\rangle = -h(X,Y) = -\langle SX, Y\rangle$.
$\langle \tilde{\nabla}_{\tilde{X}}N, N\rangle = \frac{1}{2}X\langle N, N\rangle = \frac{1}{2}X(1) = 0$.
So $\tilde{\nabla}_{\tilde{X}}N = -SX$. □

Since $\tilde{Rm} = 0$ for $\mathbb{R}^{n+1}$, the Gauss equation implies:

$$Rm(X,Y,Z,W) = h(X,W)h(Y,Z) - h(X,Z)h(Y,W)$$

If γ is a unit-speed geodesic in M, then:

$$\ddot{\gamma} = D_t\dot{\gamma} + h(\dot{\gamma},\dot{\gamma})N$$

Since it is a geodesic, $D_t\dot{\gamma} = 0$. Thus $\ddot{\gamma} = h(\dot{\gamma},\dot{\gamma})N$. So the geodesic curvature of γ is $|\ddot{\gamma}| = |h(\dot{\gamma},\dot{\gamma})|$.

$\exists$ ONB $\{E_i\}$ of T_pM so $SE_i = \kappa_i E_i$.

- κ_i: principal curvature.
- E_i: principal direction.

In this case, $h(X,Y) = \sum \kappa_i X_i Y_i$.

Definition 15.6. The **Gaussian curvature** of M is $K = \kappa_1 \dots \kappa_n$.
The **mean curvature** is $H = \frac{1}{n}(\kappa_1 + \cdots + \kappa_n)$.

16 Lecture 16

16.1 Shape Operator and Curvature of Hypersurfaces

Suppose $M \subset \mathbb{R}^{n+1}$ is a hypersurface with $\dim M = n$. Let N be a local unit normal vector field on M, and X, Y be tangent vector fields to M. Recall that on the ambient manifold $\tilde{M} = \mathbb{R}^{n+1}$, we have the Gauss formula:

$$\tilde{\nabla}_X Y = \nabla_X Y + h(X, Y)N$$

Definition 16.1. The **shape operator** S is a section of $\mathrm{End}(TM)$ defined such that:

$$\langle SX, Y \rangle = h(X, Y)$$

In local coordinates, $S^i_j = g^{ik} h_{kj}$.

Lemma 16.2. *The shape operator $S_p : T_pM \to T_pM$ is self-adjoint.*

Weingarten Equation:

$$\tilde{\nabla}_X N = -S(X)$$

Proof. For any $Y \in T_pM$, since $\langle N, Y \rangle = 0$, we differentiate it to get:

$$\langle \tilde{\nabla}_X N, Y \rangle = -\langle N, \tilde{\nabla}_X Y \rangle = -h(X, Y) = -\langle SX, Y \rangle$$

Also, since $\langle N, N \rangle = 1$, we have:

$$\langle \tilde{\nabla}_X N, N \rangle = \frac{1}{2} X \langle N, N \rangle = \frac{1}{2} X(1) = 0$$

Since $\tilde{\nabla}_X N$ has no normal component and its tangential component matches $-S(X)$, we conclude $\tilde{\nabla}_X N = -S(X)$. □

Since the ambient space $\mathbb{R}^{n+1}$ is flat ($\tilde{Rm} = 0$), the Gauss equation yields:

$$Rm(X, Y, Z, W) = h(X, W)h(Y, Z) - h(X, Z)h(Y, W)$$

At $p \in M$, since S is self-adjoint, there exists an orthonormal basis (ONB) $\{E_i\}$ of T_pM such that $SE_i = k_i E_i$.

- k_i: **principal curvature**
- E_i: **principal direction**

In this basis, $h(X, Y) = \sum k_i X_i Y_i$.

Definition 16.3. The **Gaussian curvature** of M is $K(p) = k_1 \dots k_n$.
The **mean curvature** of M is $H(p) = \frac{1}{n}(k_1 + \cdots + k_n)$.

Remark 16.4. Note that if we change the choice of normal field $N \to -N$, then $h \to -h$, and $k_i \to -k_i$. Consequently, $K \to (-1)^n K$ and $H \to -H$.

Let γ be a curve in M. Its acceleration in $\mathbb{R}^{n+1}$ is:

$$\ddot{\gamma} = D_t \dot{\gamma} + h(\dot{\gamma}, \dot{\gamma})N$$

If γ is a unit-speed geodesic in M, then $D_t \dot{\gamma} = 0$. So $\ddot{\gamma} = h(\dot{\gamma}, \dot{\gamma})N$. Thus, the geodesic curvature of γ is $|\ddot{\gamma}| = |h(\dot{\gamma}, \dot{\gamma})|$.

Examples

Example 16.5. Rigid motion to M so that $0 \in M$ and $T_0M \cong \mathbb{R}^n \subset \mathbb{R}^{n+1}$. Locally, $M = \text{graph}(f)$ where $f : \mathbb{R}^n \to \mathbb{R}$, $f(0) = 0$, and $\nabla f(0) = 0$. Consider a curve $\gamma(t) = (v^1 t, \dots, v^n t, f(vt))$.

$$\dot{\gamma}(t) = \Big(v^1, \dots, v^n, \sum v^i (\partial_i f)(vt)\Big)$$
$$\ddot{\gamma}(t) = \Big(0, \dots, 0, \sum v^i v^j (\partial_i \partial_j f)(vt)\Big)$$

At $t = 0$: $\dot{\gamma}(0) = (\vec{v}, 0)$ and $\ddot{\gamma}(0) = (0, \dots, 0, \sum v^i v^j (\partial_i \partial_j f)(0))$. Let the normal vector be $N = (0, \dots, 0, 1)$ at $p = 0$. Then $\langle \ddot{\gamma}(0), N\rangle = \sum v^i v^j (\partial_i \partial_j f)(0) = v^T (\nabla^2 f)(0) v$. By the acceleration formula, $\langle \ddot{\gamma}, N\rangle = h(\dot{\gamma}, \dot{\gamma})$. Thus, $h(\vec{v}, \vec{v}) = v^T (\nabla^2 f)(0) v$. So the principal curvatures $\{k_i\}$ are exactly the eigenvalues of the Hessian $(\nabla^2 f)(0)$.

Example 16.6. Let $M \subset \mathbb{R}^3$, $\dim M = 2$. Near the origin, $f = \frac{1}{2}k_1 x^2 + \frac{1}{2}k_2 y^2$. If $k_1, k_2 > 0$: parabolic surface. If $k_1 k_2 < 0$: saddle surface.

Example 16.7. $S^2 \subset \mathbb{R}^3$. A geodesic is $\gamma(t) = (R\cos(\frac{t}{R}), R\sin(\frac{t}{R}), 0)$. Let N be the inward normal. $\dot{\gamma} = (-\sin(\frac{t}{R}), \cos(\frac{t}{R}), 0)$, and $\ddot{\gamma} = -\frac{1}{R^2}\gamma = \frac{1}{R}N$. The shape operator is $S_p = \frac{1}{R}\,\mathrm{Id}_{T_pM}$. Here $k_1 = k_2 = \frac{1}{R}$, $K = \frac{1}{R^2}$, and $H = \frac{1}{R}$.

16.2 Theorema Egregium and Sectional Curvature

Theorem 16.8 (Gauss's Theorema Egregium)**.** *If $M^2 \subset \mathbb{R}^3$, then the Gaussian curvature K_p satisfies*

$$K_p = \frac{Rm(X, Y, Y, X)}{|X|^2|Y|^2 - \langle X, Y\rangle^2}$$

for any basis $\{X, Y\}$ of T_pM.

Proof. Recall the Gauss equation: $Rm(X, Y, Z, W) = h(X, W)h(Y, Z) - h(X, Z)h(Y, W)$. Choose $\{e_1, e_2\}$ as an ONB of T_pM consisting of principal directions. Then:

$$Rm(e_1, e_2, e_2, e_1) = h(e_1, e_1)h(e_2, e_2) - h(e_1, e_2)h(e_2, e_1) = k_1 k_2 - 0 = \det(S_p) = K$$

Hence $K_p = \frac{Rm(e_1,e_2,e_2,e_1)}{|e_1|^2|e_2|^2 - \langle e_1, e_2\rangle^2}$. It can be checked by multilinearity that the expression $\frac{Rm(X,Y,Y,X)}{|X|^2|Y|^2 - \langle X,Y\rangle^2}$ is independent of the choice of basis. $\square$

Because the right-hand side depends only on the intrinsic metric g and its Levi-Civita connection (via Rm), we can define the Gaussian curvature of any abstract 2D Riemannian manifold as:

$$K(p) = \frac{Rm(X, Y, Y, X)}{|X|^2|Y|^2 - \langle X, Y\rangle^2}$$

Proposition 16.9. *For a 2D Riemannian manifold:*

1. $Rm(X, Y, Z, W) = K(p)\big(\langle X, W\rangle\langle Y, Z\rangle - \langle X, Z\rangle\langle Y, W\rangle\big)$
2. $Ric(X, Y) = K(p)\langle X, Y\rangle$
3. $S(p) = 2K(p)$

Proof. 1. Since $\{e_1, e_2\}$ is an ONB, writing X, Y, Z, W in terms of $\{e_1, e_2\}$ and using multilinearity proves the first identity.
2. The Ricci tensor is the trace of the Riemann tensor. Let $Ric_{il} = g^{jk} Rm_{ijkl}$. Using part 1:

$$\begin{aligned} Ric_{il} &= K(p) g^{jk}(g_{il} g_{jk} - g_{ik} g_{jl}) \\ &= K(p)(g_{il}(g^{jk} g_{jk}) - g^{jk} g_{ik} g_{jl}) \\ &= K(p)(2g_{il} - \delta_i^j g_{jl}) \\ &= K(p)(2g_{il} - g_{il}) = K(p) g_{il} \end{aligned}$$

3. The scalar curvature S is the trace of the Ricci tensor:

$$S = g^{il} Ric_{il} = K(p) g^{il} g_{il} = 2K(p)$$

□

Sectional Curvature

Let $\Pi \subset T_pM$ be a 2-plane. Let $S_\Pi = \exp_p(\Pi)$ be the surface near p generated by geodesics tangent to Π.

Definition 16.10. The **sectional curvature** $K(\Pi)$ is defined as the Gaussian curvature of S_Π at p.

Proposition 16.11. *If X, Y span Π, then $K(\Pi) = \frac{Rm(X,Y,Y,X)}{|X|^2|Y|^2 - \langle X,Y\rangle^2}$.*

Proof. Claim: S_Π has vanishing 2nd fundamental form II at p. For any $v \in \Pi$, γ_v is a radial geodesic in M and also in S_Π. By the corollary of the Gauss formula:

$$0 = \tilde{D}_t \dot{\gamma} = D_t \dot{\gamma} + \mathrm{II}(\dot{\gamma}, \dot{\gamma})$$

Evaluating at p (i.e., $t = 0$), we get $D_t \dot{\gamma}(0) = 0$ and $\mathrm{II}_p(v, v) = 0$. Hence $\mathrm{II}_p = 0$. By the Gauss equation, $\tilde{Rm}^{S_\Pi} = Rm^M$ at p. Therefore, $K(\Pi) = \frac{Rm^{S_\Pi}(X,Y,Y,X)}{|X|^2|Y|^2 - \langle X,Y\rangle^2} = \frac{Rm^M(X,Y,Y,X)}{|X|^2|Y|^2 - \langle X,Y\rangle^2}$. □

Lemma 16.12. *Given T_pM and g_p, the sectional curvature at p completely determines the Riemann curvature tensor $Rm(\cdot,\cdot,\cdot,\cdot)$ at p.*

Proof. Omitted. □

Say v is a unit vector in T_pM. Let $\{e_i\}$ be an ONB of T_pM with $v = e_1$.

$$Ric(v, v) = \sum_j Rm(e_1, e_j, e_j, e_1) = \sum_j K(e_1 \wedge e_j)$$

i.e., $Ric(v, v) = (n-1) \times$ (Average sectional curvature at p of 2-planes in T_pM containing v).

$$S = \sum_{i,j} Rm(e_i, e_j, e_j, e_i) = n(n-1) \times \text{(Average sectional curvature at } p).$$

Example 16.13. $S^n \subset \mathbb{R}^{n+1}$ with radius R. The second fundamental form is $h = \frac{1}{R} g$ w.r.t the inward normal.

$$Rm(X, Y, Z, W) = h(X, W)h(Y, Z) - h(X, Z)h(Y, W) = \frac{1}{R^2}(g(X, W)g(Y, Z) - g(X, Z)g(Y, W))$$

$\Rightarrow$ Constant sectional curvature $\frac{1}{R^2}$.

Example 16.14. Hyperboloid $H^n \subset \mathbb{R}^{1,n}$ of radius R. $\Rightarrow$ Constant sectional curvature $-\frac{1}{R^2}$.

16.3 Relation Between Curvature and Geodesics via Jacobi Fields

Let $\gamma : [a, b] \to M$ be a smooth curve. Let $\Gamma : (-\varepsilon, \varepsilon) \times [a, b] \to M$ be a smooth variation of γ. We have already shown the symmetry lemma: $D_t \partial_s \Gamma = D_s \partial_t \Gamma$.

Lemma 16.15. *If W is a vector field along Γ, then:*

$$D_s D_t W - D_t D_s W = R(\partial_s, \partial_t) W$$

Suppose for each $s \in (-\varepsilon, \varepsilon)$, the curve $\gamma_s(t) = \Gamma(s, t)$ is a geodesic. Then $D_t \partial_t \Gamma = 0$ for all s. Applying D_s to this equation gives $D_s D_t \partial_t \Gamma = 0$. By the lemma:

$$D_t D_s \partial_t \Gamma + R(\partial_s, \partial_t) \partial_t \Gamma = 0$$

Using the symmetry lemma ($D_s \partial_t \Gamma = D_t \partial_s \Gamma$):

$$D_t D_t \partial_s \Gamma + R(\partial_s, \partial_t) \partial_t \Gamma = 0$$

Put $J(t) = \frac{\partial \Gamma}{\partial s}(0, t)$. Then at $s = 0$, we get the **Jacobi equation**:

$$D_t D_t J + R(J, \dot{\gamma}) \dot{\gamma} = 0$$

Any solution to the Jacobi equation is called a **Jacobi field** along γ.

Proposition 16.16. *Given an interval $I \subset \mathbb{R}$, $a \in I$, and a geodesic $\gamma : I \to M$. For any given $X, Y \in T_{\gamma(a)} M$, there is a unique Jacobi field J such that $J(a) = X$ and $(D_t J)(a) = Y$.*

Proof. Let $\{E_i\}$ be an ONB at $p = \gamma(a)$, and extend it to a parallel basis $\{E_i(t)\}$ along γ. Write $J(t) = J^l(t) E_l(t)$. Plugging this into the Jacobi equation gives:

$$(J^l)'' E_l + R(J^i E_i, \dot{\gamma}) \dot{\gamma} = 0$$

In components:

$$(J^l)'' + R^l_{ijk} J^i \dot{\gamma}^j \dot{\gamma}^k = 0$$

This is a second-order linear ODE system. Given initial data $\{J^l(a), (J^l)'(a)\}$, it has a unique smooth solution. □

Corollary 16.17. *The set of all Jacobi fields along a geodesic is a vector space of dimension $2n$.*

Proposition 16.18. *If J_1 and J_2 are Jacobi fields along γ, then the quantity*

$$\langle D_t J_1, J_2 \rangle - \langle J_1, D_t J_2 \rangle$$

is constant in t.

Proof. Take the derivative with respect to t:

$$\begin{aligned}
\partial_t(\langle D_t J_1, J_2 \rangle - \langle J_1, D_t J_2 \rangle) &= \langle D_t D_t J_1, J_2 \rangle + \langle D_t J_1, D_t J_2 \rangle - \langle D_t J_1, D_t J_2 \rangle - \langle J_1, D_t D_t J_2 \rangle \\
&= \langle D_t D_t J_1, J_2 \rangle - \langle J_1, D_t D_t J_2 \rangle \\
&= \langle -R(J_1, \dot{\gamma}) \dot{\gamma}, J_2 \rangle - \langle J_1, -R(J_2, \dot{\gamma}) \dot{\gamma} \rangle \\
&= -Rm(J_1, \dot{\gamma}, \dot{\gamma}, J_2) + Rm(J_2, \dot{\gamma}, \dot{\gamma}, J_1)
\end{aligned}$$

By the symmetries of the Riemann curvature tensor, $Rm(J_1, \dot{\gamma}, \dot{\gamma}, J_2) = Rm(J_2, \dot{\gamma}, \dot{\gamma}, J_1)$. Thus the derivative is 0, meaning the quantity is constant. □

Proposition 16.19. *The form defined by $\omega(J_1, J_2) = \langle D_t J_1, J_2\rangle - \langle J_1, D_t J_2\rangle$ makes the space of Jacobi fields along γ into a symplectic vector space.*

Proof. We have to show that the form is non-degenerate: if $\omega(J_1, J_2) = 0$ for all J_2, then $J_1 = 0$. Fix $t = a$. Pick the specific Jacobi field J_2 subject to the initial conditions:

$$J_2(a) = (D_t J_1)(a), \quad (D_t J_2)(a) = -J_1(a)$$

Since $\omega(J_1, J_2)$ is constant, evaluating it at $t = a$ gives:

$$\begin{aligned}\omega(J_1, J_2) &= \langle D_t J_1(a), J_2(a)\rangle - \langle J_1(a), D_t J_2(a)\rangle \\ &= \langle D_t J_1(a), D_t J_1(a)\rangle - \langle J_1(a), -J_1(a)\rangle \\ &= |D_t J_1(a)|^2 + |J_1(a)|^2\end{aligned}$$

If $\omega(J_1, J_2) = 0$, then $|D_t J_1(a)|^2 + |J_1(a)|^2 = 0$. This implies $J_1(a) = 0$ and $D_t J_1(a) = 0$. By the uniqueness of the solution to the Jacobi equation with zero initial data, we conclude that $J_1 \equiv 0$. $\square$

17 Lecture 17

17.1 Jacobi Fields

Proposition 17.1. *Say $\gamma : I \to M$. If M is complete or I is compact, then any Jacobi field along γ is the variation field of some variation of γ by geodesics.*

Example 17.2. In $\mathbb{R}^n$, $\gamma_s(t) = \vec{\gamma}(t) + s(\vec{X} + (t-a)\vec{Y})$.

Proof. Given J, suppose $0 \in I$. Write $p = \gamma(0)$, $v = \gamma'(0)$, so $\gamma(t) = \exp_p(tv)$. Choose a curve $\sigma : (-\epsilon, \epsilon) \to M$ and a smooth vector field $V(s)$ along σ so that $\sigma(0) = p$, $V(0) = v$, $\sigma'(0) = J(0)$, and $(D_sV)(0) = (D_tJ)(0)$.

Put $\Gamma(s,t) = \exp_{\sigma(s)}(tV(s))$. Put $W(t) = \frac{\partial \Gamma}{\partial s}(0,t)$, which is a Jacobi field along γ.

$$W(0) = \left.\frac{\partial}{\partial s}\right|_{s=0} \Gamma(s,0) = \left.\frac{d}{ds}\right|_{s=0} \sigma(s) = \sigma'(0) = J(0)$$

$$(D_tW)(0) = (D_t\partial_s\Gamma)(0,0) = (D_s\partial_t\Gamma)(0,0) = (D_sV)(0) = (D_tJ)(0)$$

Since W and J are Jacobi fields with the same initial conditions, $W = J$. □

"Trivial" Jacobi fields: $J_0(t) = \gamma'(t)$ and $J_1(t) = t\gamma'(t)$. Both are tangent to γ.

Definition 17.3. A vector field $V(t)$ along γ is **normal** if $\langle V(t), \dot{\gamma}(t)\rangle = 0$ for all t.

Proposition 17.4. *The following are equivalent (TFAE) for a Jacobi field J along γ:*

1. *J is a normal vector field.*
2. *J is orthogonal to γ at two points.*
3. *J and D_tJ are orthogonal to γ at one point.*

Proof. Put $f(t) = \langle J(t), \dot{\gamma}(t)\rangle$. Then $f'(t) = \langle D_tJ, \dot{\gamma}\rangle$.

$$f''(t) = \langle D_t^2 J, \dot{\gamma}\rangle = \langle -R(J,\dot{\gamma})\dot{\gamma}, \dot{\gamma}\rangle = 0$$

i.e., f is an affine function of t. Then: $1 \iff f = 0$. $2 \iff f(t_1) = f(t_2) = 0$. $3 \iff f(t_1) = f'(t_1) = 0$. Since an affine function with two zeros or a double zero is identically zero, the conditions are equivalent. □

Corollary 17.5. $\dim\{$*Normal Jacobi fields along* $\gamma\} = 2n-2$.

Proof. Consider the map $\{\text{Jacobi fields along } \gamma\} \to T_{\gamma(a)}M \oplus T_{\gamma(a)}M$ given by $J \mapsto (J(a), (D_tJ)(a))$. Define a map $T_{\gamma(a)}M \oplus T_{\gamma(a)}M \xrightarrow{\alpha} \mathbb{R} \oplus \mathbb{R}$ by $(V,W) \mapsto (\langle V, \gamma'(a)\rangle, \langle W, \gamma'(a)\rangle)$. α is surjective. So $\ker(\alpha) \cong \{\text{Normal Jacobi fields}\}$ has dimension $(2n) - 2 = 2n-2$. □

Lemma 17.6. *If $\gamma : I \to M$ is a geodesic, and J is a Jacobi field along γ with $J(0) = 0$. Then J is the variation field of $\Gamma(s,t) = \exp_p(t(v+sw))$, where $p = \gamma(0)$, $v = \gamma'(0)$, and $w = (D_tJ)(0)$.*

Proof. Take $\sigma(s) = p$ for all s. $V(s) = v + sw \in T_pM$ is a vector field along σ. $(D_sV)(0) = w = (D_tJ)(0)$. Using the previous proof, the variation field of Γ is exactly J. □

Proposition 17.7. *If $\gamma(t) = \exp_p(tv)$, then $J(t) = (d\exp_p)_{tv}(tw)$.*

Proof. Put $\Gamma(s,t) = \exp_p(t(v+sw))$. (For fixed s, this is a geodesic in t).

$$J(t) = \left.\frac{\partial \Gamma}{\partial s}\right|_{(0,t)} = \left.\frac{d}{ds}\right|_{s=0} \exp_p(tv + stw) = (d\exp_p)_{tv}(tw)$$

□

17.2 Constant Sectional Curvature and Polar Coordinates

Recall the Jacobi equation:

$$D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$$

Suppose M has constant sectional curvature c, and $|\dot{\gamma}| = 1$. Look for solutions with $J(0) = 0$, $\langle J, \dot{\gamma}\rangle = 0$.

$$R(X,Y)Z = c(\langle Y, Z\rangle X - \langle X, Z\rangle Y) \implies R(J,\dot{\gamma})\dot{\gamma} = c(\langle \dot{\gamma}, \dot{\gamma}\rangle J - \langle J, \dot{\gamma}\rangle\dot{\gamma}) = cJ$$

Thus, the Jacobi equation becomes:

$$D_t^2 J + cJ = 0$$

Say $E(t)$ is a parallel normal vector field along γ. Try $J(t) = f(t)E(t)$. We get $f'' + cf = 0$, which implies:

$$f(t) = \begin{cases} t & \text{if } c = 0 \\ R\sin(t/R) & \text{if } c = 1/R^2 \\ R\sinh(t/R) & \text{if } c = -1/R^2 \end{cases}$$

Corollary 17.8. *If M has constant sectional curvature c, then the metric in polar normal coordinates is:*

$$g = \begin{cases} dr^2 + r^2 g_{S^{n-1}} & \textit{if } c = 0 \\ dr^2 + R^2\sin^2(r/R) g_{S^{n-1}} & \textit{if } c = 1/R^2 \\ dr^2 + R^2\sinh^2(r/R) g_{S^{n-1}} & \textit{if } c = -1/R^2 \end{cases}$$

Proof. We can write $g = dr^2 + h_{ij}(r,\theta)d\theta^i d\theta^j$. Given $\sum \tau^i|_{t=1}$, we get a Jacobi field $J = \sum \tau^i \frac{\partial}{\partial\theta_i}$.

$$|J|^2 = \sum h_{ij}c^i c^j = \text{const} \cdot f(r)^2 \quad (\text{where } f(r) = r, R\sin(r/R), R\sinh(r/R))$$

For some constant matrix M_{ij}. As $r \to 0$, $h_{ij}(r,\theta)d\theta^i d\theta^j \sim r^2 g_{S^{n-1}}$ (since the geometry goes Euclidean). So $M_{ij}d\theta^i d\theta^j = g_{S^{n-1}}$, hence $h_{ij}(r,\theta)d\theta^i d\theta^j = f(r)^2 g_{S^{n-1}}$. □

17.3 Conjugate Points

Definition 17.9. If γ is a geodesic from p to q, we say q is **conjugate** to p along γ if there exists a (non-trivial) Jacobi field J so that $J(p) = 0$ and $J(q) = 0$.

Example 17.10. In S^2, let p be the north pole and q be the south pole.

Proposition 17.11. $\exp_p$ *is a local diffeomorphism at $v \in T_pM$ if and only if* $\exp_p(v)$ *is not conjugate to p along $\gamma(t) = \exp_p(tv)$.*

Proof. $\exp_p$ is not a local diffeomorphism at $v \iff d\exp_p$ is singular at v. $\iff (d\exp_p)_v w = 0$ for some $w \in T_pM$, $w \neq 0$. $\iff \exists$ normal Jacobi field J with $J(0) = 0$, $(D_t J)(0) = w$, and $J(q) = 0$, where $q = \exp_p(v)$. $\iff \exists$ Jacobi field J so $J(p) = 0$, $J(q) = 0$. □

Recall: If J vanishes at 2 points, then J is a normal Jacobi field.

Proposition 17.12. *If M has non-positive sectional curvature ($K \leq 0$), then M has no conjugate points.*

Proof. Suppose γ is a geodesic, J is a Jacobi field along γ, and $J(a) = J(b) = 0$. We have $D_t^2 J + R(J, \dot\gamma)\dot\gamma = 0$. Taking the inner product with J and integrating from a to b:

$$\begin{aligned}
0 &= \int_a^b \langle D_t^2 J + R(J,\dot\gamma)\dot\gamma, J\rangle \, dt \\
&= \int_a^b \left[\frac{d}{dt}\langle D_t J, J\rangle - \langle D_t J, D_t J\rangle + \langle R(J,\dot\gamma)\dot\gamma, J\rangle\right] dt \\
&= \langle D_t J, J\rangle\Big|_a^b + \int_a^b \left(-|D_t J|^2 + \langle R(J,\dot\gamma)\dot\gamma, J\rangle\right) dt \\
&= 0 - \int_a^b |D_t J|^2 \, dt - \int_a^b \langle R(J,\dot\gamma)J, \dot\gamma\rangle \, dt \\
&= -\int_a^b |D_t J|^2 \, dt - \int_a^b K(J \wedge \dot\gamma)\left(|J|^2|\dot\gamma|^2 - \langle J, \dot\gamma\rangle^2\right) dt
\end{aligned}$$

Since $K \le 0$, the second integral is ≤ 0 (its negation is ≥ 0). Thus the entire expression is ≤ 0. This forces $\int_a^b |D_t J|^2 \, dt = 0 \implies D_t J = 0$. But $J(a) = 0$, so $J \equiv 0$. $\square$

Theorem 17.13 (Cartan-Hadamard)**.** *If M is connected, complete, with non-positive sectional curvatures, then $\exp_p : T_pM \to M$ is a covering map.*

18 Lecture 18

18.1 The Cartan-Hadamard Theorem

Theorem 18.1 (Cartan-Hadamard)**.** *If M is complete, connected, and $K \leq 0$. Then for any $p \in M$, $\exp_p : T_pM \to M$ is a covering map.*

Proof. $\exp_p : T_pM \to M$ is defined on all of T_pM.

Put $\tilde{g} = \exp_p^* g$, a Riemannian metric on T_pM. $\exp_p : (T_pM, \tilde{g}) \to (M, g)$ is a local isometry. Since straight lines through $0 \in T_pM$ correspond to geodesics through p in M, from previous results, $\exp_p$ is a covering map. □

Corollary 18.2. *If M is connected, complete and has $K \leq 0$, then $\pi_i(M) = 0$ for $i > 1$. i.e., any continuous map $f : S^i \to M$ can be continuously deformed to a point map.*

Proof. Since $T_pM \cong \mathbb{R}^n$ is simply connected and contractible, any map $f : S^i \to M$ can be lifted to a map $F : S^i \to T_pM$ such that $\exp_p \circ F = f$. Since $T_pM \cong \mathbb{R}^n$, there exists a continuous deformation (homotopy) F_s of $F = F_0$ to a point map F_1. Put $f_s = \exp_p \circ F_s$. This provides the continuous deformation of f to a point. □

Example 18.3. If M is diffeomorphic to S^n $(n > 1)$, then M has no metric with $K \leq 0$, since $\pi_n(S^n) = \mathbb{Z} \neq 0$.
Remark / Cor: Any compact manifold of dim > 2 has a metric with negative Ricci curvature, i.e., $Ric(v, v) < 0$ for all $v \neq 0$.

Corollary 18.4. *If $K \leq 0$, then all of the topology of M comes from $\pi_1(M)$.*

18.2 2nd Variation of Length

Theorem 18.5. *Say $\gamma : [a, b] \to M$ is a unit speed geodesic. Γ is a proper variation of γ, and V is the variation field. Then*

$$\left.\frac{d^2}{ds^2}\right|_{s=0} L(\gamma_s) = \int_a^b |D_t V^\perp|^2 - \langle R(V^\perp, \dot{\gamma})\dot{\gamma}, V^\perp\rangle \, dt.$$

Proof. First assume Γ is smooth. The length functional is:

$$L(\gamma_s) = \int_a^b \left\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle^{1/2} dt.$$

Taking the first derivative with respect to s:

$$\begin{aligned}\frac{d}{ds} L(\gamma_s) &= \int_a^b \left\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle^{-1/2} \left\langle \frac{\partial \Gamma}{\partial t}, D_s \frac{\partial \Gamma}{\partial t} \right\rangle dt \\ &= \int_a^b \left\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle^{-1/2} \left\langle \frac{\partial \Gamma}{\partial t}, D_t \frac{\partial \Gamma}{\partial s} \right\rangle dt.\end{aligned}$$

Taking the second derivative with respect to s:

$$\begin{aligned}\frac{d^2}{ds^2} L(\gamma_s) = \int_a^b & -\frac{1}{2} \left\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle^{-3/2} 2 \left\langle \frac{\partial \Gamma}{\partial t}, D_t \frac{\partial \Gamma}{\partial s} \right\rangle \left\langle \frac{\partial \Gamma}{\partial t}, D_t \frac{\partial \Gamma}{\partial s} \right\rangle \\ & + \left\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle^{-1/2} \left\langle D_s D_t \frac{\partial \Gamma}{\partial s}, \frac{\partial \Gamma}{\partial t} \right\rangle + \left\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle^{-1/2} \left\langle D_t \frac{\partial \Gamma}{\partial s}, D_t \frac{\partial \Gamma}{\partial s} \right\rangle dt.\end{aligned}$$

Evaluate at $s = 0$. We have $\frac{\partial \Gamma}{\partial t} = \dot{\gamma}$, $\frac{\partial \Gamma}{\partial s} = V$, and since γ is unit speed, $\langle \dot{\gamma}, \dot{\gamma} \rangle = 1$. Also recall that $D_s D_t \frac{\partial \Gamma}{\partial s} = D_t D_s \frac{\partial \Gamma}{\partial s} + R(V, \dot{\gamma})V$. Thus:

$$\frac{d^2}{ds^2}\bigg|_{s=0} L(\gamma_s) = \int_a^b -\langle \dot{\gamma}, D_t V \rangle^2 + |D_t V|^2 + \langle \dot{\gamma}, R(V, \dot{\gamma})V \rangle dt + \langle \dot{\gamma}, D_s \frac{\partial \Gamma}{\partial s} \rangle \bigg|_a^b.$$

(Note that $\int \langle \dot{\gamma}, D_t D_s V \rangle dt = \int \frac{d}{dt} \langle \dot{\gamma}, D_s V \rangle dt$ since $D_t \dot{\gamma} = 0$, giving the boundary term).

Decompose V:

$$V = \langle V, \dot{\gamma} \rangle \dot{\gamma} + V^\perp \implies D_t V = \langle D_t V, \dot{\gamma} \rangle \dot{\gamma} + D_t V^\perp$$

From this, $|D_t V|^2 - \langle D_t V, \dot{\gamma} \rangle^2 = |D_t V^\perp|^2$.
Also, by symmetries of the Riemann curvature tensor:

$$\langle \dot{\gamma}, R(V, \dot{\gamma})V \rangle = -\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = -\langle R(V^\perp, \dot{\gamma})\dot{\gamma}, V^\perp \rangle.$$

For a proper variation, the boundary terms vanish, leading to:

$$\implies \frac{d^2}{ds^2}\bigg|_{s=0} L(\gamma_s) = \int_a^b |D_t V^\perp|^2 - \langle R(V^\perp, \dot{\gamma})\dot{\gamma}, V^\perp \rangle \, dt.$$

□

Definition 18.6. The **index form** is defined as:

$$I(V, W) = \int_a^b \langle D_t V, D_t W \rangle - \langle R(V, \dot{\gamma})\dot{\gamma}, W \rangle \, dt.$$

For proper normal vector fields $V^\perp$, $I(V^\perp, V^\perp) = \frac{d^2}{ds^2}\Big|_{s=0} L(\gamma_s)$.

Corollary 18.7. *If γ is a minimal curve from p to q, then $I(V, V) \geq 0$ for all proper normal vector fields V.*

Proposition 18.8. *If γ_1 and γ_2 are unit speed minimal geodesics from p to q, then neither is minimizing beyond q.*

Proposition 18.9. *γ cannot be minimizing beyond a conjugate point.*

Proof. Say $\gamma(a)$ and $\gamma(c)$ are conjugate, where $a < c < b$. We want to show γ is not minimizing from a to b. Since $\gamma(c)$ is conjugate to $\gamma(a)$, $\exists$ a nonzero Jacobi field J so that $J(a) = 0$ and $J(c) = 0$.

Take $V(t) = \begin{cases} J(t) & a \leq t \leq c \\ 0 & c \leq t \leq b \end{cases}$

$I(V, V) = I(J, J) + 0$. However, by integration by parts:

$$I(J, J) = -\int_a^c \langle D_t^2 J + R(J, \dot{\gamma})\dot{\gamma}, J \rangle \, dt + \sum_i \langle \Delta D_t V, V(t_i) \rangle$$

Since J is a Jacobi field, the integral is zero. Since $J(a) = J(c) = 0$, the boundary term vanishes. So $I(V, V) = 0$.

Take W so that $W(c) = -(D_t J)(c^-) \neq 0$. Choose W to be a proper variation field. Let $X = V + \varepsilon W$. Then:

$$I(X, X) = I(V, V) + 2\varepsilon I(V, W) + \varepsilon^2 I(W, W) = 2\varepsilon I(V, W) + \varepsilon^2 I(W, W).$$

Evaluate $I(V, W)$:

$$I(V, W) = -\int_a^b \langle D_t^2 V + R(V, \dot{\gamma})\dot{\gamma}, W\rangle \, dt + \sum \langle \Delta D_t V, W(c)\rangle$$

The integral is 0 because V is a Jacobi field on $[a, c]$ and zero on $[c, b]$. The jump in the derivative at $t = c$ is:

$$\Delta D_t V = D_t V(c^+) - D_t V(c^-) = 0 - D_t J(c^-) = -D_t J(c^-).$$

So, $I(V, W) = \langle -D_t J(c^-), W(c)\rangle = \langle -D_t J(c^-), -D_t J(c^-)\rangle = -|D_t J(c^-)|^2 \neq 0$. Hence:

$$I(X, X) = -2\varepsilon |D_t J(c^-)|^2 + \varepsilon^2 I(W, W).$$

For small $\varepsilon > 0$, $I(X, X) < 0$. This forces γ to not be minimizing. $\square$

Definition 18.10. The **cut point** of p along γ_v is $\gamma_v(t_{\text{cut}})$, where

$$t_{\text{cut}}(v) = \sup\{b > 0 : \gamma_v|_{[0,b]} \text{ is minimizing}\}.$$

Definition 18.11. The **cut locus** of p is $\{q \in M : q \text{ is a cut point of } p \text{ along some geodesic}\}$.

19 Lecture 19

19.1 Classical Theorems

Proposition 19.1. *If γ_1, γ_2 are two distinct minimizing geodesics from p to q, then neither is minimizing beyond q.*

Proposition 19.2. *γ cannot be minimizing beyond a conjugate point.*

Theorem 19.3 (Bonnet)**.** *Suppose M is a connected complete Riemannian manifold, with sectional curvatures $K \geq \frac{1}{R^2} > 0$. Then M is compact,* $\operatorname{diam}(M) = \sup_{p,q \in M} d(p,q) \leq \pi R$, *and $\pi_1(M)$ is finite.*

Proof. Suppose $\operatorname{diam}(M) > \pi R$, i.e., $\exists p, q \in M$, and a minimizing geodesic $\gamma : p \to q$, defined on $[0, L] \to M$ with $L > \pi R$.

Let E be a parallel unit normal vector field along γ. Put $V(t) = \sin\left(\frac{\pi t}{L}\right) E(t)$, which is a proper normal vector field along γ. We have $D_t V = \left(\frac{\pi}{L} \cos\left(\frac{\pi t}{L}\right)\right) E(t)$, and the index form is:

$$\begin{aligned}
I(V,V) &= \int_0^L \left[|D_t V|^2 - R(V, \dot\gamma, \dot\gamma, V) \right] dt \\
&= \int_0^L \left[\frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \sin^2\left(\frac{\pi t}{L}\right) K(\dot\gamma \wedge E) \right] dt \\
&\leq \int_0^L \left[\frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \frac{1}{R^2} \sin^2\left(\frac{\pi t}{L}\right) \right] dt \\
&= \frac{L}{2} \left[\frac{\pi^2}{L^2} - \frac{1}{R^2} \right] < 0
\end{aligned}$$

since $L > \pi R \implies \frac{\pi^2}{L^2} < \frac{1}{R^2}$. This contradicts the fact that γ is length-minimizing (which requires $I(V,V) \geq 0$).

Hence, $\operatorname{diam}(M) \leq \pi R$. Thus, for any $p \in M$, $M = \exp_p\left(\overline{B_{\pi R}(0)}\right)$. Since $\exp_p$ is continuous and the closed ball $\overline{B_{\pi R}(0)} \subset T_p M$ is compact, M is compact.

Look at the universal cover $\tilde{M}$. It also has $K \geq \frac{1}{R^2}$. Apply the same argument to $\tilde{M}$, so $\tilde{M}$ is compact. Hence $\pi_1(M)$ is finite. □

Theorem 19.4 (Myers)**.** *The same conclusion holds if the Ricci curvature satisfies* $\operatorname{Ric}(v,v) \geq \frac{n-1}{R^2} g(v,v)$ *for all $v \in TM$.*

Proof. Suppose $\operatorname{diam}(M) > \pi R$. Find a minimizing geodesic $\gamma : [0, L] \to M$ from p to q with $L > \pi R$.

Let $\{E_i\}$ be an orthonormal basis of $\dot\gamma^\perp \in T_{\gamma(t)} M$. Extend these to be parallel vector fields $E_i(t)$ along γ. Put $V_i(t) = \sin\left(\frac{\pi t}{L}\right) E_i(t)$ for each $1 \leq i \leq n-1$.

$$I(V_i, V_i) = \int_0^L \left[\frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \sin^2\left(\frac{\pi t}{L}\right) R(E_i, \dot\gamma, \dot\gamma, E_i) \right] dt$$

Summing over all i:

$$\begin{aligned}
\sum_{i=1}^{n-1} I(V_i, V_i) &= \int_0^L \left[(n-1) \frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \sin^2\left(\frac{\pi t}{L}\right) \operatorname{Ric}(\dot\gamma, \dot\gamma) \right] dt \\
&\leq \int_0^L \left[(n-1) \frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \sin^2\left(\frac{\pi t}{L}\right) \frac{n-1}{R^2} \right] dt \\
&= (n-1) \frac{L}{2} \left[\frac{\pi^2}{L^2} - \frac{1}{R^2} \right] < 0
\end{aligned}$$

This implies that for at least one i, $I(V_i, V_i) < 0$, which is a contradiction. Hence $\operatorname{diam}(M) \leq \pi R$. The rest of the proof follows exactly as in Bonnet's Theorem. $\square$

Remark 19.5. Positive curvature $\implies$ focusing geodesics.
Negative curvature $\implies$ defocusing geodesics.

Theorem 19.6 (Synge)**.** *Suppose M is compact and connected with strictly positive sectional curvature ($K > 0$).*

1. *If n is even and M is orientable, then M is simply connected.*
2. *If n is odd, then M is orientable.*

Proof of 1. First, we state a lemma from linear algebra:

Lemma 19.7. *If $A \in SO(k)$ and k is odd, then $\exists v \neq 0$ such that $Av = v$.*

Proof of Lemma:

$$\begin{aligned}\det(A - I) &= \det(A^{-1}) \cdot \det(A - I) = \det(I - A^{-1}) \\ &= \det(I - A^T) = \det((I - A)^T) = \det(I - A) \\ &= (-1)^k \det(A - I) = -\det(A - I) \quad (\text{since } k \text{ is odd})\end{aligned}$$

This implies $\det(A - I) = 0$, so 1 is an eigenvalue.

Now, suppose n is even, M is orientable, and $\pi_1(M) \neq \{0\}$. We know there is a bijection:

$$\{\text{Free homotopy classes of loops}\} \longleftrightarrow \{\text{Conjugacy classes of } \pi_1(M)\}$$

Since $\pi_1(M) \neq 0$, there exists a non-trivial homotopy class of loops, so we can find a non-trivial closed geodesic γ that minimizes length in this class.

Take p on γ. Look at parallel transport around γ, yielding an isometry $P_\gamma \in \operatorname{End}(T_pM)$. Write $P_\gamma = I|_{\langle \dot\gamma_p \rangle} \oplus A$, where $A = P_\gamma|_{\dot\gamma_p^\perp}$. Choose an orthonormal basis for $\dot\gamma_p^\perp$. Since M is orientable, parallel transport preserves orientation, so $A \in SO(n-1)$. Because n is even, $n - 1$ is odd. By the Lemma, $\exists v \in \dot\gamma_p^\perp$ with $Av = v$. Say $|v| = 1$.

Let $V(t)$ be the vector field along γ obtained by parallel transport of v. Because $Av = v$, V smoothly closes up and defines a global parallel normal vector field along the closed geodesic γ. The index form for V is:

$$I(V, V) = \int_0^L \left[|D_tV|^2 - R(V, \dot\gamma, \dot\gamma, V)\right] dt = -\int_0^L K(V, \dot\gamma) dt < 0$$

since $D_tV = 0$ and $K > 0$. This implies we can deform γ in the direction of V to strictly decrease its length, which is a contradiction since γ minimizes length in its free homotopy class. $\square$

Corollary 19.8. $\mathbb{RP}^2 \times \mathbb{RP}^2$ *has no metric with positive sectional curvature.*

Remark 19.9. **Hopf's Conjecture:** $S^2 \times S^2$ has no metric with strictly positive sectional curvature.

19.2 The Cut Locus

Suppose (M, g) is connected and complete. Let $p \in M$ and let $\gamma_v(t) = \exp_p(tv)$ be a unit-speed geodesic (i.e., $|v| = 1$).

Definition 19.10. Let $t(v) := \sup\{b > 0 \mid \gamma_v|_{[0,b]} \text{ is minimizing}\} \in (0, \infty]$.
The **cut point** of p along γ_v is defined as $\gamma_v(t(v))$.
The **cut locus** of p is the set $\{q \in M \mid q \text{ is a cut point of } p \text{ along some geodesic}\}$.

Proposition 19.11. *Given $p \in M$, $v \in T_pM$ with $|v| = 1$. Put $c = t(v)$.*

1. *If $0 < b < c$, then $\gamma_v|_{[0,b]}$ has no conjugate points and is the unique minimizing geodesic between its endpoints.*

2. *If $c < \infty$, then $\gamma_v|_{[0,c]}$ is minimizing, and either $\gamma_v(c)$ is conjugate to p along γ_v, OR there exist at least two distinct minimizing geodesics from p to $\gamma_v(c)$.*

Proof. **Part 1:** Choose $b < b' < c$. If $\gamma_v|_{[0,b']}$ had a conjugate point, or if there existed two distinct minimizing geodesics to $\gamma_v(b')$, then γ_v would cease to be minimizing beyond b'. Thus $\gamma_v|_{[0,b']}$ would not be minimizing, which contradicts the definition of $c = t(v)$.

Part 2: To show γ_v is minimizing on $[0, c]$: take $b_i \nearrow c$. Then $d(p, \gamma_v(b_i)) = b_i$. By continuity of the distance function, $d(p, \gamma_v(c)) = \lim d(p, \gamma_v(b_i)) = \lim b_i = c$.

Assume $c < \infty$. We want to show the alternative conditions. Take $b_i \searrow c$. Since $b_i > c$, $\gamma_v|_{[0,b_i]}$ is not minimizing. However, M is complete, so there exists a unit-speed minimizing geodesic $\sigma_i : [0, a_i] \to M$ such that $\sigma_i(a_i) = \gamma_v(b_i)$, where $a_i = d(p, \gamma_v(b_i)) < b_i$.

Put $w_i = \sigma_i'(0) \in T_pM$, with $|w_i| = 1$. Since the unit sphere in T_pM is compact, after passing to a subsequence, we can assume $w_i \to w$ for some unit vector $w \in T_pM$, and $a_i \to a$. By continuity, $\sigma_i(a_i) = \exp_p(a_i w_i) \to \exp_p(aw) = \gamma_w(a)$. On the other hand, $\sigma_i(a_i) = \gamma_v(b_i) \to \gamma_v(c)$. So $\exp_p(aw) = \gamma_v(c)$.

Also, the distance is continuous:

$$c = d(p, \gamma_v(c)) = \lim d(p, \sigma_i(a_i)) = \lim a_i = a$$

So $a = c$. The geodesic $\sigma : [0, c] \to M$ given by $\sigma(t) = \exp_p(tw)$ is also a minimizing curve from p to $\gamma_v(c)$.

- If $v \neq w$, then σ and γ_v are two distinct minimizing geodesics from p to $\gamma_v(c)$.

- If $v = w$, we need to show $\gamma_v(c)$ is conjugate to p. For large i, we have $a_i w_i \neq b_i v$ (since $|w_i| = |v| = 1$ and $a_i < b_i$). However, we know $\exp_p(a_i w_i) = \sigma_i(a_i) = \gamma_v(b_i) = \exp_p(b_i v)$. We also have $a_i w_i \to cw = cv$ and $b_i v \to cv$. Since $\exp_p$ maps distinct sequences of points converging to cv to the same values, $\exp_p$ is not injective in any neighborhood of $cv \in T_pM$. By the Inverse Function Theorem, the differential $(d\exp_p)_{cv}$ cannot be invertible. Therefore, cv is a critical point of $\exp_p$, which by definition means $\gamma_v(c)$ is conjugate to p along γ_v.

$\square$

20 Lecture 20

20.1 Injectivity Domain and Injectivity Radius

Definition 20.1. The **injectivity domain** of p, denoted $ID(p) \subset T_pM$, is defined as:

$$ID(p) = \{v \in T_pM : |v| < t_{\text{cut}}(v/|v|)\}$$

Proposition 20.2. $\exp_p$ *is a diffeomorphism of* $ID(p)$ *to its image* $M \setminus cut(p)$.

Proof. There are no conjugate points in $ID(p)$, so $\exp_p$ is a local diffeomorphism on $ID(p)$. Moreover, $\exp_p$ is injective on $ID(p)$, since if two geodesics from p meet, they cannot be minimizing beyond that intersection point. Hence, the image is exactly $M \setminus \text{cut}(p)$. □

Definition 20.3. The **injectivity radius** at p is:

$\text{inj}(p) = \sup\{a > 0 : \exp_p$ is a diffeomorphism from $B_a(0) \subset T_pM$ to its image in $M\} \in \mathbb{R}^+ \cup \{\infty\}$.

Proposition 20.4. *If* M *is complete,* $\text{inj}(p) = d(p, cut(p))$.

Proof. Put $d = d(p, \text{cut}(p))$. Given a, we write B_a for an open ball with radius a around $0 \in T_pM$. We claim that $\exp_p$ is a local diffeomorphism $B_a \to \exp_p(B_a)$ if and only if $a \leq d$.

- Suppose $a \leq d$, then $B_a \subset ID(p)$, and we are done.
- If $a > d$, then there exists a cut point q for p with $d(p, q) < a$. Take the radial geodesic from p to q; it is not minimizing beyond q since q is a cut point. So there is no normal ball of radius $> d(p, q)$ that $\exp_p$ is a local diffeomorphism on it.

□

Example 20.5. Metrics on S^2.

Proposition 20.6. *If* M *is diffeomorphic to* S^2*, then* $cut(p)$ *cannot contain a loop.*

Proof. Suppose L is a loop in $\text{cut}(p)$. Take q inside L and take a minimizing geodesic from p to q. It must intersect L, hence it cannot be minimizing beyond the intersection. □

20.2 Comparison Geometry

Recall the Jacobi equation:

$$D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$$

Bound on curvature ⇒ Bound on Jacobi field ⇒ Bound on Riemannian metric.

Model case: A scalar equation $f'' + Kf = 0$, where $K \equiv K(t)$. This is the Sturm-Liouville equation:

$$(*) \quad \frac{f''}{f} + K = 0$$

Put $h = \frac{f'}{f}$ and get:

$$(**) \quad h' + h^2 + K = 0 \quad \text{(Riccati equation)}$$

Definition 20.7. If $U \subset M$, a **local distance function** on U is a smooth function f with $|\nabla f| = 1$.

Put $F = \nabla f$. We previously showed that $\nabla_F F = 0$. Every flow line of F is a unit-speed geodesic.

Definition 20.8. Let $\mathcal{H} \in \Gamma(T_1^1(U))$ be given by $\mathcal{H}(W) = \nabla_W F$ for any $W \in TU$. Note that $\mathcal{H} \in \Gamma(\text{End}(TM))$, and U is a level set of f.

Lemma 20.9. *$\mathcal{H}(F) = 0$ and $\mathcal{H}$ is self-adjoint.*

Proof. $\mathcal{H}(F) = \nabla_F F = 0$. For the self-adjointness:

$$\langle V, \mathcal{H}(W)\rangle = \langle V, \nabla_W F\rangle = \mathrm{Hess}(f)(V, W) = \langle \mathcal{H}(V), W\rangle$$

□

Proposition 20.10.

$$\nabla_F \mathcal{H} + \mathcal{H}^2 + R_F = 0$$

where $R_F(W) = R(W, F)F$.

Proof. We compute $R(W, F)F$:

$$\begin{aligned} R(W, F)F &= \nabla_W \nabla_F F - \nabla_F \nabla_W F - \nabla_{[W,F]} F \\ &= 0 - \nabla_F(\mathcal{H}(W)) - \mathcal{H}([W, F]) \quad (\text{since } \nabla_F F = 0) \\ &= -(\nabla_F \mathcal{H})(W) - \mathcal{H}(\nabla_F W) - \mathcal{H}([W, F]) \end{aligned}$$

Since $[W, F] = \nabla_W F - \nabla_F W$, we have $\nabla_F W + [W, F] = \nabla_W F$. Therefore:

$$\begin{aligned} R(W, F)F &= -(\nabla_F \mathcal{H})(W) - \mathcal{H}(\nabla_F W + [W, F]) \\ &= -(\nabla_F \mathcal{H})(W) - \mathcal{H}(\nabla_W F) \\ &= -(\nabla_F \mathcal{H})(W) - \mathcal{H}(\mathcal{H}(W)) \\ &= -(\nabla_F \mathcal{H})(W) - \mathcal{H}^2(W) \end{aligned}$$

So:

$$(\nabla_F \mathcal{H})(W) + \mathcal{H}^2(W) + R(W, F)F = 0 \iff \nabla_F \mathcal{H} + \mathcal{H}^2 + R_F = 0$$

□

Corollary 20.11. *On a flowline γ of F,*

$$D_t \mathcal{H} + \mathcal{H}^2 + R_{\dot\gamma} = 0$$

where $R_{\dot\gamma}(W) = R(W, \dot\gamma)\dot\gamma$.

Geometry meaning: $F = \nabla f$ is a unit normal to a level set of f. Hence $\nabla_W F$ is the shape operator of this level set, with respect to $N = -\nabla f$. (If W is tangent to the level set, this is the Weingarten map).

Main example: Let $p \in M$, and let $U = (\text{A normal neighborhood of } p) \setminus \{p\}$. Let f be the radial function r, i.e., $f(q) = d(p, q)$. By Gauss's lemma, $F = \nabla f$ is exactly ∂_r.

Constant curvature c: Put

$$s_c(r) = \begin{cases} r & \text{if } c = 0 \\ R\sin(r/R) & \text{if } c = \frac{1}{R^2} \\ R\sinh(r/R) & \text{if } c = -\frac{1}{R^2} \end{cases}$$

The metric can be written as $g = dr^2 + s_c^2(r) g_{S^{n-1}}$.

By $SO(n-1)$ symmetry, $\mathcal{H} = \alpha(r)\Pi_r$, where $\Pi_r : T_qM \to T_qM$ is the projection onto $\partial_r^\perp$. We have the equation $\alpha' + \alpha^2 + c = 0$, with the asymptotic behavior $\alpha(r) \sim \frac{1}{r}$ as $r \to 0$.

We can find that $\alpha(r) = \frac{s_c'(r)}{s_c(r)}$, and check:

$$\alpha' + \alpha^2 + c = \frac{s_c'' s_c - (s_c')^2}{s_c^2} + \left(\frac{s_c'}{s_c}\right)^2 + c = \frac{s_c''}{s_c} + c = 0$$

(since s_c satisfies $s_c'' + c s_c = 0$).

20.3 Matrix Riccati Equation Comparison

Let V be an inner product space, and A, B be endomorphisms of V.

Definition 20.12. We define $A \geq 0$ if $\langle v, Av \rangle \geq 0$ for all $v \in V$. We write $A \geq B$ if $A - B \geq 0$. Let $S(n, \mathbb{R})$ denote the space of symmetric $n \times n$ matrices.

Theorem 20.13. *Say $H, \tilde{H} : [a, b] \to S(n, \mathbb{R})$ satisfy the Riccati equations:*

$$H' + H^2 + S = 0$$
$$\tilde{H}' + \tilde{H}^2 + \tilde{S} = 0$$

where $S, \tilde{S} : [a, b] \to S(n, \mathbb{R})$ are continuous, and $\tilde{S}(t) \geq S(t)$ for all $t \in [a, b]$.

Suppose $\lim_{t \to a}(\tilde{H}(t) - H(t)) \leq 0$. Then $\tilde{H}(t) - H(t) \leq 0$ for all $t \in [a, b]$.